IMPLEMENTATION OF A WEB SOLUTION FOR HOUSEHOLD ITEM RENTALS

BY:

AJAYI PRINCE AYOKUNLE

MATRIC NUMBER: 215221

SUPERVISED BY DR. ADERONKE B. SAKPERE

DEPARTMENT OF COMPUTER SCIENCE, FACULTY OF SCIENCE,

UNIVERSITY OF IBADAN

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF A BACHELOR OF SCIENCE DEGREE IN COMPUTER SCIENCE (B.Sc. COMPUTER SCIENCE)

MARCH 2025.

CERTIFICATION

This is to certify that the project titled **"Implementation of a Web Solution for Household Item Rentals"** was carried out by **Ajayi Prince Ayokunle (Matriculation Number: 215221)**.

DR. ADERONKE B. SAKPERE …….

Supervisor Date and Signature

PROF. A.B. ADEYEMO. …….

Head, Computer Science Department. Date and Signature

DEDICATION

This project is dedicated to God Almighty for His constant guidance and support throughout the completion of this work. It is also dedicated to my parents, Mr. and Mrs. Ajayi, for their unwavering love, encouragement, and support in all my endeavors.

ACKNOWLEDGEMENT

I express my deepest gratitude to God Almighty for His constant guidance, strength, and love, which enabled the successful completion of this project. His support has been my anchor throughout this journey.

I am immensely grateful to my project supervisor, Dr. Aderonke B. Sakpere, for her unwavering patience, insightful guidance, and invaluable support, all of which were instrumental in shaping this work.

I would also like to thank my parents, Mr. and Mrs. Ajayi, for their endless love, encouragement, and unwavering belief in me. Your support has been my greatest source of motivation.

I also thank my siblings, Victoria, Elizabeth, Queen, and Precious, for always being there to listen, discuss my ideas, and offer thoughtful suggestions that have helped improve this project.

Lastly, I appreciate all the friends and colleagues who supported me in one way or another during this period. Your encouragement has meant the world to me.

TABLE OF CONTENTS

CERTIFICATION i
DEDICATION ii
ACKNOWLEDGEMENT iii
TABLE OF CONTENTS iv
LIST OF FIGURES vii
LIST OF TABLES ix
ABSTRACT x
CHAPTER ONE: INTRODUCTION 1
1.1 Background: 1
1.2 Problem Statement: 4
1.3 Aim and Objectives: 5
1.4 Scope and Limitations: 6
1.5 Significance of the Study: 8
1.5.1 Impact on People and Technological Advancements 9
1.5.2 Contribution to Sustainable Development Goals (SDGs) 10
CHAPTER TWO: LITERATURE REVIEW 12
2.1 Introduction: 12
2.2 Peer-to-Peer (P2P) in the Sharing Economy 12
2.3 Other Forms of Sharing or Item Rental 13
2.4 Literature Review: 13
2.4.1 What's Mine Is Yours: The Rise of Collaborative Consumption 14
2.4.2 Putting the Sharing Economy into Perspective 14
2.4.3 The Sharing Economy: Behavioral Insights and Participation 14
2.4.4 Trust and Security in Digital Peer-to-Peer Rental Platforms 15
2.4.5 Renting Household Items in Finland 15
2.4.6 Development of Rental Application using Prototyping Method (Riyanti et al., 2024) 16
2.4.7 An User-Friendly Android-based Application for Online Rental System (B. Dhanalaxmi et al., 2021) 16
2.4.8 Web-Based Online Service Rental System (N.T. Thamara Sugandhi, 2017) 16

2.4.9 Online Car Rental System Using Web-Based and SMS Technology (Mohd Nizam Osman et al., 2017) 17
2.5 Comparative Analysis of Related Works and Systems 17
2.5.1 Airbnb (https://www.airbnb.com) and the Evolution of Short-Term Rentals 17
2.5.2 RentMyItems (https://www.rentmyitems.com) and Niche Rental Platforms 17
2.5.3 Traditional Rental Services: Cort (https://www.cort.com) and Aaron's (https://www.aarons.com) 18
2.5.4 Online Car Rental Systems as Analogous Models 18
2.6 Theoretical Framework and Conceptual Model 18
2.6.1 Core Theoretical Models 18
2.6.2 Conceptual Model for HiRentals 19
2.7 Summary 19
CHAPTER THREE: METHODOLOGY 21
3.1 Introduction: 21
3.2 System Analysis: 22
3.2.1 Understanding the Problem 22
3.2.2 Research Insights: The Need for the System 23
3.2.3 Current System Evaluation 25
3.2.4 Key Limitations of Existing Systems 26
3.2.5 Requirements Gathering 26
3.2.6 Stakeholder Input 28
3.3 System Design: 31
3.3.1 Use-Case Diagram 31
3.3.2 System Architecture 32
3.3.3 User Interface 34
3.3.4 Activity Diagrams 36
3.3.5 Sequence Diagrams 38
3.3.6 Data Flow Diagram 43
3.3.7 Entity-Relationship (ER) Model 46
3.3.8 Class Diagram 50
3.4 Summary: 52
CHAPTER FOUR: IMPLEMENTATION AND TESTING 53
4.1 Introduction: 53

4.2 Implementation: 53
4.2.1 Technology Stack 54
4.2.2 Home Page 55
4.2.3 Item Listing Page 56
4.2.4 Item Request Form 57
4.2.5 Item Details Page 58
4.2.6 Owner Dashboard 60
4.2.7 Renter Dashboard 62
4.2.8 Admin Dashboard 63
4.2.9 Authentication Pages 64
4.2.10 Database and File Storage 66
4.2.11 Integration of Calendly 66
4.2.12 Payment Processing with Paystack 67
4.3 Implementation Issues Addressed: 69
4.4 System Testing: 69
4.4.1 Unit Testing 69
4.4.2 Integration Testing 70
4.4.3 End-to-End (E2E) Testing 71
4.4.4 Performance Testing 72
4.4.5 Security Testing 73
4.4.6 Usability Testing 74
4.4.7 Usability Analysis 80
4.5 Summary: 80
CHAPTER FIVE: SUMMARY, CONCLUSION, AND RECOMMENDATIONS 83
5.1 Introduction: 83
5.2 Summary: 83
5.3 Conclusions: 84
5.4 Recommendations: 85
REFERENCES 87
APPENDIX 89

LIST OF FIGURES

Figure 3.1: Survey statistic 1 23
Figure 3.2: Survey statistic 2 24
Figure 3.3: Survey statistic 3 24
Figure 3.4: Survey statistic 4 29
Figure 3.5: Survey statistic 5 29
Figure 3.6: Use-Case diagram 32
Figure 3.7: High level System Architecture 33
Figure 3.8: UI - Homepage (Paper Prototype) 34
Figure 3.9: UI - Item list page (Paper Prototype) 35
Figure 3.10: UI - Item detail page (Paper Prototype) 36
Figure 3.11: Activity diagram - Item Request 37
Figure 3.12: Activity diagram - Item Upload 38
Figure 3.13: Sequence diagram - Item Request and Matching 39
Figure 3.14: Sequence diagram - Item Upload 40
Figure 3.15: Sequence diagram - Rent Initiation 41
Figure 3.16: Sequence diagram - Item Verification on Rent Order 42
Figure 3.17: Sequence diagram - Item Verification before Return 43
Figure 3.18: Context Diagram (Level 0 DFD) 44
Figure 3.19: Level 1 Data Flow Diagram 45
Figure 3.20: Entity-Relationship diagram 49
Figure 3.21: Class diagram 51
Figure 4.1: QR Code for Live Application 54
Figure 4.2: QR Code for Application's Source Code 54
Figure 4.3: Homepage 55
Figure 4.4: Item listing page 1 56
Figure 4.5 Item listing page 2 57
Figure 4.6: Item Request Form 58
Figure 4.7: Item details page 1 59
Figure 4.8: Item details page 2 60
Figure 4.9: Owner dashboard 1 61

Figure 4.10: Owner dashboard 2 62
Figure 4.11: Renter dashboard 63
Figure 4.12: Admin dashboard 64
Figure 4.13: Sign in page 65
Figure 4.14: Sign up page 65
Figure 4.15: Firestore Database 66
Figure 4.16: Calendly Integration 1 67
Figure 4.17: Calendly integration 2 67
Figure 4.18: Paystack Integration 68

LIST OF TABLES

Table 4.1: Modules Tested and Sample Test Cases 69
Table 4.2: Integration Test Summary 70
Table 4.3: End-to-End Testing Summary 71
Table 4.4: Performance Testing Summary 73
Table 4.5: Security Test Summary 74
Table 4.6: Demographic Profile of Test Participants 75
Table 4.7: Task Assignment and Success Criteria 76
Table 4.8: System Usability Scale (SUS) Scores of Participants 77
Table 4.9: Usability Test Metrics Summary 79

ABSTRACT

The project “Implementation of a Web Solution for Household Item Rentals” addresses the growing need for accessible and sustainable rental services among short-term residents in urban Nigeria. With increasing mobility among students, temporary workers, and NYSC participants, the demand for essential household items has surged, yet traditional acquisition methods remain impractical and costly.

The core problem lies in the financial and logistical challenges associated with purchasing or transporting household items for temporary stays. Existing rental services do not adequately cater to the short-term needs of these users, resulting in underutilized resources and unnecessary expenses. This project, therefore, targets the gap in the market by proposing a dedicated, peer-to-peer rental platform that simplifies access to household items.

A comprehensive development methodology was adopted to build the HiRentals platform. The system was developed using React.js with TypeScript for a responsive frontend, while Firebase Firestore and Firebase Storage ensured real-time data management and secure media handling. Additionally, the integration of Paystack for secure payment processing and Calendly for automated scheduling bolstered the system’s reliability. Rigorous testing phases—including unit, integration, end-to-end, performance, security, and usability tests—were conducted to validate the platform’s robustness and user-friendliness.

The final solution demonstrates high usability, operational efficiency, and secure transaction management, effectively connecting item owners with renters. The platform has proven to reduce financial burdens and logistical challenges while promoting sustainable consumption. Although there are minor limitations due to dependency on third-party services, the HiRentals system offers a scalable and cost-effective alternative for short-term household item acquisition, with promising avenues for future enhancements such as AI-driven verification and native mobile applications.

CHAPTER ONE:
INTRODUCTION

1.1 BACKGROUND:

Relocating for short-term periods, whether for educational, work-related, or other purposes, presents unique challenges, particularly when it comes to managing household items. The process of moving can be stressful and complicated, especially when the duration of the relocation is uncertain or temporary (Owen, 2020). For individuals such as students, temporary workers, and National Youth Service Corps (NYSC) participants, the need to acquire essential household items for short-term stays often leads to significant financial and logistical burdens. Traditional solutions, such as purchasing new items or transporting bulky belongings, are often impractical and costly, particularly for individuals with limited financial resources (Frenken & Schor, 2019).

The rise of the sharing economy, facilitated by advancements in computing and information technology, has provided innovative solutions to many of these challenges. The sharing economy, which emphasizes the sharing of underutilized resources, has gained significant traction in recent years, with platforms like Airbnb and RentMyItems revolutionizing how people access goods and services (Hamari et al., 2016). These platforms leverage digital technologies to connect individuals who have resources to share with those who need them, thereby optimizing resource utilization and reducing waste (Frenken & Schor, 2019). However, while these platforms have successfully addressed needs such as short-term accommodation and tool rentals, there remains a significant gap in the market for household item rentals, particularly in developing countries like Nigeria.

In Nigeria, the increasing mobility of individuals, particularly students and NYSC participants, has highlighted the need for a more efficient and cost-effective solution for accessing household items. For example, a student relocating to a new city for a few years may face the dilemma of either purchasing new household items or incurring high transportation costs to move their belongings from their hometown. Both options are financially burdensome and inefficient, especially when the items are only needed for a short period (Owen, 2020). Similarly, NYSC

participants deployed to rural areas often struggle to acquire basic household items, leading to suboptimal living conditions and reduced quality of life.

Recent advancements in computing and information technology have created opportunities to address these challenges through innovative digital platforms. The proliferation of mobile devices, cloud computing, and real-time data management systems has enabled the development of peer-to-peer (P2P) rental platforms that can connect users seamlessly and efficiently (Dhanalaxmi et al., 2021). Cloud-based databases, for instance, allow for real-time data synchronization, enabling users to access up-to-date information on available items and rental transactions. Additionally, secure payment gateways facilitate seamless and secure financial transactions, enhancing user trust and confidence in the platform (Sugandhi, 2017).

The integration of scheduling tools further enhances the efficiency of rental platforms by automating the process of arranging meetings for item verification and handover. This not only reduces the administrative burden on users but also ensures that transactions are conducted transparently and securely (Riyanti et al., 2024). Moreover, the use of responsive web design frameworks ensures that platforms are accessible across a wide range of devices, including smartphones, tablets, and desktops, thereby increasing their reach and usability (Zervas et al., 2017).

Despite these technological advancements, there remains a lack of specialized platforms that cater specifically to the needs of short-term residents in developing countries. Existing rental platforms, such as Airbnb and RentMyItems, either focus on specific niches like accommodation or lack the robust verification and trust-building mechanisms necessary for household item rentals (Cheng, 2020). This gap in the market underscores the need for a dedicated platform that leverages modern computing technologies to provide a seamless, secure, and cost-effective solution for short-term household item rentals.

In summary, the challenges faced by short-term residents in acquiring household items highlight the need for innovative solutions that leverage advancements in computing and information technology. By developing a peer-to-peer rental platform specifically designed for household items, this project aims to address the financial and logistical burdens faced by individuals in temporary living situations. The platform will utilize modern technologies such as cloud-based databases, secure payment gateways, and automated scheduling tools to provide a user-friendly and efficient solution that promotes resource sharing and sustainability.

1.2 PROBLEM STATEMENT:

Many individuals, particularly students, temporary workers, and mobile professionals, face significant challenges when relocating for short-term stays. These challenges include the inconvenience and high costs associated with owning, transporting, or acquiring household items for temporary accommodations. Whether it is the financial burden of purchasing new items or the impracticality of transporting large household goods across long distances, these issues create logistical and financial difficulties for short-term residents.

In many cases, individuals are forced to choose between purchasing items they will only need for a brief period or incurring substantial transportation costs to move their belongings. This not only increases their living expenses but also leads to inefficient resource use, as items may be left unused for extended periods.

There is a growing need for a more efficient and cost-effective solution that allows short-term residents to access household items on demand, without the burden of ownership or expensive logistics. A flexible, peer-to-peer rental platform designed to meet the specific needs of short-term occupants could offer a viable solution to this problem. By enabling easy access to essential household items through rentals, individuals can avoid unnecessary costs and inconveniences, while fostering a more sustainable and cost-effective way of living in temporary accommodations.

1.3 AIM AND OBJECTIVES:

Aim

The aim of this project is to develop a peer-to-peer web system that allows short-term occupants to easily and securely rent household items, while fostering trust and transparency between users.

Objectives

1. To design and implement a peer-to-peer matching system that connects renters and lenders based on location, budget, and rental duration, ensuring a seamless experience for both parties.
2. To design a trust-building mechanism by introducing vetted verification agents who are responsible for inspecting and confirming the condition of items before and after rentals, thereby ensuring reliability and security.
3. To design and implement a secure and efficient payment system that tracks transactions and ensures timely payments between renters and lenders, enhancing user confidence in the platform.

1.4 SCOPE AND LIMITATIONS:

Scope

This project focuses on the design and implementation of a web-based peer-to-peer rental system designed specifically for short-term residents. The platform aims to connect renters with lenders of household items, allowing users to rent essential items such as refrigerators, mattresses, gas cookers, and other household goods. The system will feature a matching module based on location, budget, and rental duration, a trust-building mechanism using verification agents or admins, and a secure payment system for transactions.

The study will cover:

- The design and development of the platform's core features (user registration, item listing, item upload, item search, and booking).
- The implementation of a verification system for items, ensuring both renters and lenders have a trusted platform for transactions.
- The integration of a secure payment processing system to facilitate transactions between renters and lenders.
- The system will be designed for use in urban settings in Nigeria, where there is a high demand for short-term rentals.

Limitations

Despite the comprehensive scope of the project, there are certain limitations:

- Geographical Scope: The system will be primarily designed for use in Nigerian urban areas. Although it could be expanded to other regions, this study will not focus on international scalability or adaptation.
- User Base: The project’s primary focus is on short-term residents, such as students and temporary workers. However, it will not consider long-term rental agreements or the needs of commercial businesses.
- Technological Constraints: The web system will be designed to work on desktop and mobile browsers, but it will not include mobile applications (native apps for Android/iOS) due to time and resource constraints.

- Security Concerns: While the system will include a secure payment gateway and verification mechanism, the implementation of advanced security measures, such as end-to-end encryption or sophisticated fraud detection systems, will be limited within the scope of this study.
- Market Adoption: The study will not include a detailed analysis of market adoption rates or customer behavior, as it focuses on the development and functionality of the system itself.

These limitations are taken into account when designing the scope of the study, ensuring that while the project is comprehensive in its approach, it remains manageable within the given timeframe and resources.

1.5 SIGNIFICANCE OF THE STUDY:

The concept of rental systems has existed for many years, with a wide range of platforms available for renting various goods and services, from houses to cars and electronics. However, despite the prevalence of rental platforms, there is a significant gap in the market when it comes to household item rentals for short-term residents. Many rental platforms cater to specific or broad needs, such as real estate, cars, or general rental services. However, the ability to easily and affordably rent household items like fridges, mattresses, and cooking equipment for temporary stays remains largely unaddressed.

In Nigeria, a country with a fast-growing urban population, short-term residents, including students, workers on temporary assignments, and National Youth Service Corps (NYSC) participants, are often forced to make do with limited or inadequate living conditions due to the high costs of transporting personal belongings or purchasing household items. Short-term residents moving to urban areas, like Ibadan, Lagos, or Abuja, are faced with the dilemma of either buying new items they will only use for a brief period or incurring expensive transportation costs to bring their personal belongings from home. In many cases, the latter is neither practical nor financially feasible, leading to unnecessary financial strain.

The proposed study seeks to fill this gap by introducing a peer-to-peer web solution specifically designed for renting household items. By providing a platform that allows individuals to rent household items for the duration of their temporary stay, the system offers a cost-effective, flexible, and sustainable solution. Renters can easily access necessary household goods without the burden of ownership, while lenders can generate additional income from underutilized household items. This sharing economy model fosters an environment where resources are optimized, and the financial and logistical burdens of short-term living are minimized.

One key feature of this system is its focus on trust and security. A common challenge with peer-to-peer systems is the need to ensure that both renters and lenders can trust each other and the items being exchanged. This study addresses this issue by introducing a verification mechanism,

where vetted agents will inspect items before and after each rental, ensuring that both parties can confidently engage in transactions. The integration of a secure payment system further adds to the reliability of the platform, making it not just a transactional tool, but also a trusted service within the community.

1.5.1 Impact on People and Technological Advancements

This study is significant because it directly addresses the financial strain and logistical challenges faced by short-term residents in Nigeria. Students, workers, and other individuals who frequently relocate for short-term assignments can benefit from a system that provides easy access to household items without the need for large upfront purchases or expensive transportation. The flexibility of the platform allows users to rent items based on their specific needs, saving money and reducing waste.

For example, students who are only in a city for a few years can rent furniture, electronics, and kitchenware for the duration of their stay, without the need to buy new or second-hand items that may not be usable once they graduate. Similarly, NYSC participants deployed to locations far from home can access the essentials they need for a year-long stay without the financial burden of purchasing new items. This economic relief not only benefits individuals but also contributes to a more efficient allocation of resources, as items are reused and shared among multiple users, reducing unnecessary consumption.

On a broader scale, this system also aligns with technological advancements in the sharing economy. As more industries turn to digital platforms to facilitate peer-to-peer exchanges, the proposed system leverages web-based technology to provide a seamless and user-friendly experience. With an easy-to-use interface, users can list items for rent, browse available items, make secure transactions, and communicate directly with other users. The inclusion of location-based search features further enhances the platform’s accessibility, making it a versatile solution for a wide range of users in urban areas.

1.5.2 Contribution to Sustainable Development Goals (SDGs)

The proposed system contributes directly to several key Sustainable Development Goals (SDGs), including:

- **SDG 8**: Decent Work and Economic Growth

  By creating an avenue for individuals to rent out household items, the system generates additional income streams for those with underutilized goods. This not only provides economic opportunities for lenders but also supports local economies by circulating goods within communities, promoting economic growth and financial stability.

- **SDG 11**: Sustainable Cities and Communities

  The system fosters more sustainable living practices by encouraging resource sharing, reducing waste, and promoting the reuse of household items. This aligns with SDG 11, which aims to make cities and human settlements more inclusive, safe, resilient, and sustainable. By optimizing the use of resources in urban centers, the system contributes to creating more sustainable communities.

- **SDG 12**: Responsible Consumption and Production

  The rental platform promotes responsible consumption by reducing the need for individuals to purchase new items for short-term use. Instead, people can rent items that they will only need temporarily, minimizing unnecessary production and consumption. This contributes to the reduction of waste and supports circular economy principles.

- **SDG 1**: No Poverty

  The opportunity for individuals to earn income by renting out household items can serve as an additional source of income, especially for those who may not have the means to invest in traditional business ventures. By giving people the ability to rent out unused goods, the system supports efforts to alleviate financial pressures, thereby contributing to poverty reduction.

In summary, this project is significant not only because it addresses the pressing challenges faced by short-term residents in Nigeria but also because it aligns with global sustainability goals and technological advancements. By offering an affordable, efficient, and sustainable solution for renting household items, this system can have a meaningful impact on the lives of its users. Additionally, its contribution to income generation, resource optimization, and community trust will foster a more connected and sustainable urban ecosystem. This project is not just a solution to a specific problem; it is a step towards a more collaborative and sustainable way of living in urban areas.

CHAPTER TWO:
LITERATURE REVIEW

2.1 INTRODUCTION:

The rapid evolution of digital technologies has given rise to the sharing economy, a paradigm that challenges traditional notions of ownership by emphasizing access over possession. In recent years, this model has transformed how individuals acquire, utilize, and share goods and services, particularly in domains such as temporary household item rentals. For short-term residents such as students, NYSC participants, and temporary workers, the cost and logistical burdens of purchasing and transporting household items have created a pressing need for efficient, peer-to-peer (P2P) rental platforms. This literature review examines seminal works on collaborative consumption, explores trust and security mechanisms fundamental to P2P systems, and compares existing platforms like Airbnb and RentMyItems with traditional rental models. The review also outlines relevant theoretical frameworks that underpin user adoption and system sustainability.

2.2 PEER-TO-PEER (P2P) IN THE SHARING ECONOMY

Peer-to-peer (P2P) refers to an organizational model in which individuals directly exchange goods, services, or information among themselves rather than relying on a central authority or intermediary. This decentralized approach is often associated with digital platforms that enable participants to connect, negotiate terms, and complete transactions on their own. Early conceptualizations of P2P arose in technology contexts such as file-sharing networks (e.g., Napster), but the model has since expanded to encompass broader social and economic activities (Sundararajan, 2016). In the sharing economy, P2P typically describes consumer-to-consumer (C2C) marketplaces in which people make underused resources—such as rooms, cars, or household items—available to others for a fee or for temporary use. Belk (2014) highlights how this model democratizes access by turning consumers into both providers and recipients, effectively blurring the line between personal ownership and commercial services. Trust mechanisms, such as ratings, reviews, and secure payment gateways, play a critical role in facilitating these transactions. Although P2P models promise greater flexibility, community building, and cost efficiency, challenges remain around liability, insurance coverage, and regulatory compliance (Frenken & Schor, 2019).

2.3 Other Forms of Sharing or Item Rental

While P2P is a prominent structure in the sharing economy, there are several other forms of sharing or rental systems. One common variant is business-to-consumer (B2C) rental, where professional companies maintain ownership of assets—such as cars, bicycles, or appliances—and rent them out to consumers for short periods (Henten & Windekilde, 2016). In this model, the business usually handles maintenance, insurance, and customer service, while users benefit from predictable standards of quality and availability. Another approach is cooperative-based sharing, in which a group of individuals collectively owns and manages resources for mutual benefit (Zhang, Li, & Zhao, 2022). These cooperatives may operate at a neighborhood or community level, with participants pooling money or labor to maintain shared assets. A more recent innovation is subscription-based sharing, sometimes referred to as product-as-a-service (PaaS), where users pay a recurring fee to access items on demand (Piscicelli, Cooper, & Fisher, 2015). In this setup, companies or communities rotate inventory among subscribers, ensuring that resources circulate efficiently while minimizing idle time. Each of these models—B2C, cooperative-based, and subscription-based—presents its own balance of control, cost structure, and user autonomy. They also underscore that the broader sharing economy is not monolithic but rather a tapestry of organizational forms and motivations, ranging from purely profit-driven to more socially or environmentally oriented.

2.4 Literature Review:

This section presents a comprehensive review of key academic and empirical works that examine both the theoretical foundations and practical applications of peer-to-peer (P2P) rental systems, with a particular focus on household items. It begins by exploring seminal texts on collaborative consumption such as Botsman and Rogers (2010) and further expanded by Frenken and Schor (2019) and Hamari et al. (2016) which illustrate how shifting from ownership to access can reduce waste and optimize resource utilization. The review then delves into region-specific challenges by analyzing Siefen's (2020) study on renting household items in Finland, highlighting issues such as high transportation costs and the unique difficulties faced by transient residents in developing contexts.

Furthermore, the section addresses trust and security which are critical elements for any successful P2P platform, drawing on insights from Cheng (2020) and Sundararajan (2016).

2.4.1 What's Mine Is Yours: The Rise of Collaborative Consumption

Botsman and Rogers' seminal work, *What's Mine Is Yours: The Rise of Collaborative Consumption (2010)*, lays the groundwork for understanding the sharing economy. Their analysis explains how traditional notions of ownership are being replaced by models that emphasize access over possession. They argue that the collaborative consumption model not only reduces waste but also reallocates resources more efficiently by connecting underutilized assets with those in need. This paradigm shift has significant implications for household item rentals, where temporary access to goods can alleviate the financial and logistical burdens of ownership. By enabling short-term residents to rent essential household items, platforms modeled on this philosophy can foster sustainability and economic efficiency.

2.4.2 Putting the Sharing Economy into Perspective

Frenken and Schor (2019) provide a contemporary perspective on the sharing economy by contextualizing its environmental and societal impacts in *Putting the Sharing Economy into Perspective*. Their research highlights how digital platforms facilitate the efficient allocation of resources, reduce environmental waste, and promote sustainable consumption patterns. They underscore that the success of sharing economy platforms relies on robust digital infrastructures that ensure seamless user experiences. For household item rentals, this means that a platform must integrate real-time data management, secure transactions, and reliable verification processes to ensure both efficiency and trustworthiness. The insights from this study support the potential benefits of a dedicated rental system that addresses the unique challenges of temporary residency, particularly in developing regions.

2.4.3 The Sharing Economy: Behavioral Insights and Participation

Hamari, Sjöklint, and Ukkonen (2016) delve into the motivations driving user participation in the sharing economy in their paper, *The Sharing Economy: Why People Participate in Collaborative Consumption*. Their empirical study reveals that factors such as convenience, cost savings, and environmental consciousness play pivotal roles in user adoption. They find that trust and perceived value significantly influence participation rates, which is particularly relevant for platforms that mediate household item rentals. By offering a cost-effective alternative to ownership, a well-

designed rental platform can tap into these motivators. Moreover, the study suggests that the incorporation of user reviews, transparent pricing, and secure payment mechanisms can enhance user confidence and promote higher engagement levels.

2.4.4 Trust and Security in Digital Peer-to-Peer Rental Platforms

2.4.4.1 Trust-Building Mechanisms

Cheng (2020) in *Sharing Economy: A Review and Agenda for Future Research* emphasizes that trust is the cornerstone of any successful peer-to-peer (P2P) platform. He argues that in the absence of traditional institutional safeguards, digital platforms must rely on alternative mechanisms such as rigorous user verification, item quality checks, and comprehensive rating systems to foster trust. For a household item rental system, this translates to implementing pre- and post-rental inspections, secure identity verification, and transparent user feedback processes. These mechanisms are essential to mitigate risks and encourage user participation by creating a reliable environment where both owners and renters feel secure in their transactions.

2.4.4.2 Security Considerations

Complementing the focus on trust, Sundararajan (2016) in *The Sharing Economy: The End of Employment and the Rise of Crowd-Based Capitalism* discusses the critical importance of security in sustaining digital platforms. He argues that continuous investment in security infrastructure is necessary to protect sensitive user data and ensure secure financial transactions. For a system like HiRentals, incorporating robust authentication protocols, encrypted data transfers, and reliable payment gateways (such as Paystack) is imperative. These measures not only safeguard the platform against cyber threats but also enhance overall user confidence, thereby supporting sustained engagement and growth.

2.4.5 Renting Household Items in Finland

Siefen's study, *Renting household items in Finland (2020)*, provides a regional perspective on the challenges and benefits associated with household item rentals. Siefen investigates the high logistical and transportation costs incurred by transient residents and emphasizes the potential of specialized rental systems to alleviate both financial and psychological burdens. His analysis further demonstrates the environmental benefits of shifting from ownership to renting, supporting

a circular economy. The study draws on additional literature addressing sustainable consumption patterns and access-based economic models, highlighting the need for platforms that are tailored to local conditions while promoting environmental sustainability.

2.4.6 Development of Rental Application using Prototyping Method (Riyanti et al., 2024)

This paper focuses on the creation of a mobile rental application, using iterative prototyping to refine both user experience and core functionalities. Early versions of the app were tested with real users, and feedback loops guided improvements in navigation, feature placement, and payment integration. The authors emphasize how each prototype iteration revealed new insights about user priorities, such as the need for immediate updates on item availability, transparent cost breakdowns, and reliable security measures for transactions. By repeatedly incorporating user feedback, the final prototype demonstrated heightened user satisfaction and reduced development missteps. Although scalability and data protection remain areas for further enhancement, the study provides a blueprint for building customer-centric rental apps through continuous prototyping.

2.4.7 An User-Friendly Android-based Application for Online Rental System (B. Dhanalaxmi et al., 2021)

Dhanalaxmi and colleagues introduce an Android app designed to streamline rentals of goods and services via smartphones, reflecting the broader shift toward mobile transactions. They illustrate how a simple, intuitive interface can boost user adoption, with features such as location-based searching, push notifications, and one-tap bookings. The paper discusses backend database strategies for handling concurrent requests and ensuring item information is consistently updated. Test results show that users found the app straightforward to use, particularly when searching or comparing multiple items, and that mobile-first design can draw in audiences who might avoid more complex web portals. The authors also acknowledge that performance optimization and thorough encryption protocols are vital to maintain user trust as the platform scales.

2.4.8 Web-Based Online Service Rental System (N.T. Thamara Sugandhi, 2017)

Sugandhi's work presents a web-based approach to rental services, outlining the technical and organizational steps to migrate an often paper-driven process into a digital platform. The system's core modules revolve around listing items or services, handling user signups, and managing bookings through a centralized database. By using web technologies that prioritize user-friendly navigation and real-time data, the platform reduces operational delays and broadens access beyond

normal business hours. The paper also addresses the importance of robust security measures, including role-based permissions and input validation, to safeguard both user information and financial transactions. While the web interface expands potential clientele, the research suggests that the integration of advanced dispute resolution mechanisms and fraud detection would further enhance user confidence.

2.4.9 Online Car Rental System Using Web-Based and SMS Technology (Mohd Nizam Osman et al., 2017)

Osman and coauthors examine how combining web-based functionality with SMS notifications can enrich the user experience in car rental services. Their system allows customers to reserve vehicles online, view real-time availability, and pay securely, while simultaneously benefiting from SMS alerts about booking status, reminders before pick-up, or changes to rental terms. This dual-channel communication model aims to accommodate users who may have limited or unreliable internet access but can still receive text messages reliably. By integrating an SMS gateway, the authors demonstrate how rental agencies can cut down on missed reservations, handle cancellations more efficiently, and maintain consistent contact with customers. The study concludes that additional tools like AI-powered customer support or dynamic pricing algorithms could make the system even more responsive to market demands.

2.5 Comparative Analysis of Related Works and Systems

2.5.1 Airbnb (https://www.airbnb.com) and the Evolution of Short-Term Rentals

Airbnb is often cited as the flagship example of the sharing economy. Its success lies in its comprehensive user verification system, secure payment processing, and robust review mechanism. However, while Airbnb excels in facilitating short-term accommodation rentals, it falls short in addressing the niche of household item rentals. For instance, although some Airbnb hosts offer basic amenities, the platform does not systematically support the rental of specialized household items such as kitchen appliances or furniture. This gap creates an opportunity for a dedicated platform that focuses solely on the rental of household items—a niche that HiRentals seeks to fill.

2.5.2 RentMyItems (https://www.rentmyitems.com) and Niche Rental Platforms

RentMyItems offers a P2P rental model that allows users to rent various goods, including tools, cameras, and electronics. Despite its functional model, RentMyItems lacks the specialized focus

on household items, which requires a more nuanced approach to verification and trust-building. The limited geographic coverage and relatively simple verification processes further constrain its effectiveness in addressing the needs of short-term residents. By contrast, HiRentals proposes a tailored solution that integrates detailed verification processes and leverages local networks to enhance accessibility and trust.

2.5.3 Traditional Rental Services: Cort (https://www.cort.com) and Aaron's (https://www.aarons.com)

Traditional rental services like Cort and Aaron's provide short-term furniture and appliance rentals but often operate at higher costs and with limited flexibility. These platforms generally cater to longer-term rental needs and may not be cost-effective for temporary residents, such as students or NYSC participants. Moreover, their business models do not typically incorporate the peer-to-peer element, which can drive down costs and improve accessibility. By adopting a P2P model, HiRentals can offer a more agile and economically viable solution, better suited to the dynamic needs of short-term residents.

2.5.4 Online Car Rental Systems as Analogous Models

Research by Osman et al. (2017) on online car rental systems demonstrates that web-based solutions can effectively manage rental transactions through real-time data synchronization and secure payment gateways. Although the context differs, the underlying principles of user authentication, transaction security, and system scalability are directly applicable to household item rentals. The success factors identified in these systems provide valuable insights into designing a platform that meets both user expectations and technical requirements.

2.6 Theoretical Framework and Conceptual Model

2.6.1 Core Theoretical Models

The literature on technology adoption offers several frameworks that are pertinent to HiRentals:

- **Technology Acceptance Model (TAM):**
  TAM posits that perceived ease of use and perceived usefulness are critical determinants of user adoption. For HiRentals, ensuring a seamless and intuitive user interface, along with demonstrable cost savings and convenience, will be crucial to driving user acceptance.
- **Diffusion of Innovations Theory:**

Rogers' theory explains how innovations spread within a social system. By positioning HiRentals as an innovative solution tailored to the unique challenges of temporary residency, the platform can effectively capture early adopters and accelerate market diffusion.

- **Trust and Risk Theories:**
  Integrating insights from trust and risk management literature, it becomes evident that minimizing perceived risk through rigorous verification and secure payment systems is essential. This theoretical perspective justifies the incorporation of detailed trust-building measures in the system design.

2.6.2 Conceptual Model for HiRentals

A conceptual model for HiRentals can be visualized as follows:

- **Inputs**:
  Core system components such as the user interface, verification mechanisms, payment gateway, and scheduling system.
- **Mediators**:
  User perceptions including ease of use, trust, and perceived value.
- **Outcomes**:
  Successful user adoption, high transaction reliability, and overall platform sustainability.

This model not only guides the system's design and implementation but also provides a framework for evaluating the platform's impact on its users.

2.7 SUMMARY

The literature underscores that while the sharing economy has significantly reshaped consumer behavior in various domains, a distinct gap remains in the rental of household items for short-term residents, particularly in developing countries like Nigeria. Seminal works by Botsman and Rogers (2010), Frenken and Schor (2019), and Hamari et al. (2016) provide the theoretical and empirical underpinnings for this shift, exploring the motivations and barriers to collaborative consumption.

Siefen (2020) provides a regional analysis of household item rentals in Finland, highlighting challenges related to trust, logistics, and accessibility. Further, Cheng (2020) and Sundararajan

(2016) discuss platform-based economies and the role of digital marketplaces in driving participation. Studies by Zervas et al. (2017) and Owen (2020) offer insights into platform competition and storage alternatives, which are relevant for understanding HiRentals' positioning in the market.

Comparative analyses of platforms such as Airbnb, RentMyItems, Cort, and Aaron's reveal both the potential and limitations of existing solutions. While some platforms facilitate temporary access to goods, they often lack specialized focus on short-term household item rentals, particularly in developing economies.

Trust-building and security mechanisms emerge as critical considerations. Research by Henten and Windekilde (2016), Hamari et al. (2016), and Riyanti et al. (2024) emphasizes the need for robust verification processes, secure payment systems, and user reliability ratings to foster trust in peer-to-peer (P2P) rental models.

Additionally, the literature highlights key gaps in both market practices and academic research:

- **Lack of Specialized Focus**: Existing platforms do not cater specifically to short-term residents who need household item rentals.
- **Geographic Limitations**: Many rental services remain unavailable in developing countries like Nigeria, despite high demand for affordable, accessible solutions.
- **Trust and Verification**: P2P rental platforms often lack effective mechanisms for verifying item quality and user credibility, deterring participation.
- **Sustainability Concerns**: While some platforms promote resource sharing, they do not explicitly align with sustainable development goals (SDGs), such as reducing waste and promoting responsible consumption.

This project addresses these gaps by developing a web-based P2P rental platform designed for short-term residents in urban areas. The platform will incorporate trust-enhancing mechanisms, secure payments, and a sustainability-driven approach, ensuring it is a viable and scalable solution for the challenges faced by short-term residents in Nigeria.

CHAPTER THREE:
METHODOLOGY

3.1 INTRODUCTION:

The methodology chapter provides a detailed explanation of the processes and approaches adopted in the design and development of the web solution for household item rentals. This chapter is structured into two main sections: **System Analysis** and **System Design**, each addressing specific aspects of the project's development.

- **System Analysis**:

This section focuses on understanding the requirements of the system. It begins with findings from the requirement analysis, which includes insights gathered from surveys conducted with the target audience, such as students, NYSC participants, and other short-term residents. These findings are used to identify both the functional requirements, such as features the system must perform, and the non-functional requirements, such as system performance, scalability, and security.

- **System Design**:

  The system design section translates the requirements into visual representations and architectural plans. Various diagrams are used to illustrate the structure and functionality of the system, including:

  - Use Case Diagrams: To outline the interactions between users and the system.
  - System Architecture Diagrams: To provide a high-level view of the system's components and their relationships.
  - Entity-Relationship Diagrams (ERD): To depict the database structure and relationships between data entities.
  - Activity Diagrams: To represent workflows within the system.

  Other relevant diagrams, such as Sequence Diagrams and Data Flow Diagrams, are also included to provide comprehensive insights into the system's operation.

This structured approach ensures that the system is developed in a systematic and efficient manner, addressing both user needs and technical requirements. By following this methodology, the project

aligns with best practices in software development, delivering a solution that is both practical and innovative.

3.2 SYSTEM ANALYSIS:

The purpose of this section is to define the problem that the proposed web solution seeks to address and to gather valuable insights from potential users of the system. A thorough understanding of the problem is essential, as it establishes a solid foundation for the subsequent phases of system design and implementation.

3.2.1 Understanding the Problem

The problem that the proposed system addresses is the lack of accessible, cost-effective, and convenient solutions for temporarily renting household items. Many individuals, particularly those in short-term living situations like NYSC participants, face challenges in acquiring household items when they move into a new place temporarily. The target demographic—NYSC participants—often have to relocate for a year, making it impractical to purchase household items they may only need for a few months.

However, current solutions do not adequately solve this problem. While platforms like Airbnb facilitate temporary accommodation, they do not include the rental of essential household items (such as furniture, kitchen appliances, or electronics) necessary for temporary living. This gap in the market creates a need for a platform where users can rent household items for the short duration of their stay.

In addition, there is a lack of effective peer-to-peer rental services for household items that allow individuals to rent from others directly, ensuring a cost-effective, community-based solution. Thus, the primary objective of the proposed system is to create a peer-to-peer rental platform specifically tailored to temporary living situations, allowing users to rent out household items for short periods.

3.2.2 Research Insights: The Need for the System

The insights gathered from interviews and surveys conducted, primarily with NYSC participants and other short-stay occupants, shed light on the pressing nature of the problem:

- Financial Inconvenience: Respondents emphasized that purchasing household items for short-term use poses a significant financial burden.



Figure 3.1: Survey statistic 1

Figure 3.1 shows the survey statistic for a particular question

- Logistical Challenges: Many highlighted the difficulty of finding, transporting, and setting up essential items when relocating temporarily.

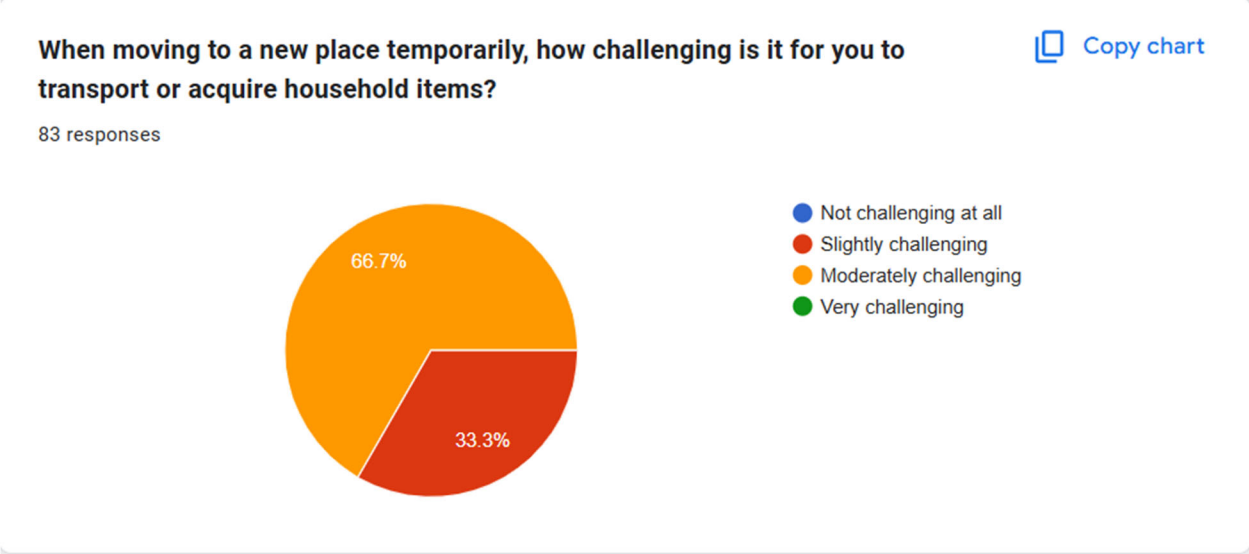


Figure 3.2: Survey statistic 2

Figure 3.2 shows the survey statistic for a particular question

- Preference for Renting: A clear preference emerged for renting over buying, especially for items that are only needed for a limited period.

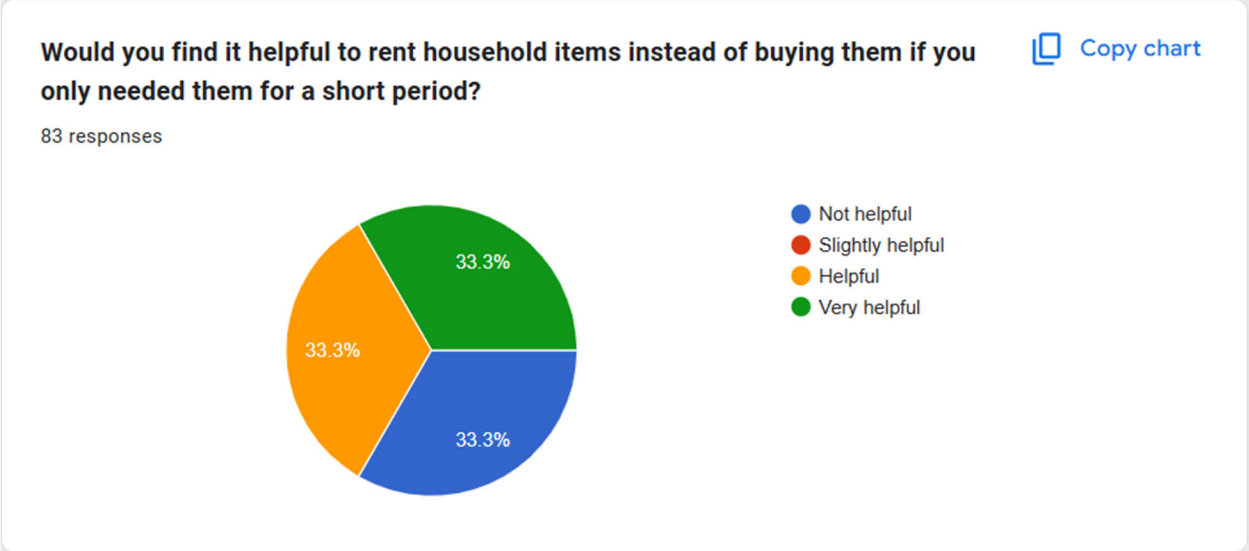


Figure 3.3: Survey statistic 3

Figure 3.3 shows the survey statistic for a particular question

The data from the research strongly indicate an unmet demand for a platform that offers a convenient, secure, and cost-effective means for individuals to rent household items temporarily.

3.2.3 Current System Evaluation

Before proposing a new solution, it is essential to evaluate existing systems to understand their strengths, weaknesses, and limitations. This evaluation helps identify what aspects of the current market can be leveraged and where improvements are necessary.

Existing Solutions

Airbnb: While Airbnb has revolutionized short-term accommodation, it does not cater to the rental of household items. Airbnb hosts may provide basic furnishings, but items such as kitchen appliances, furniture, and electronics are generally not included. For guests staying for extended periods, the lack of necessary household items becomes a considerable inconvenience.

- Strengths: Trusted platform for short-term accommodation.
- Weaknesses: Does not address the need for temporary rentals of household items.

Furniture and Appliance Rental Services: Several platforms provide short-term rentals for items like furniture or appliances, but they are often geographically limited and do not fully address the needs of temporary residents, such as those requiring flexibility and affordability.

- Strengths: Offers rental options for specific items like furniture and electronics, providing some level of convenience.
- Weaknesses: Often unavailable in certain locations, and the services are frequently expensive and impractical for short-term needs.

Peer-to-Peer Rental Platforms (e.g., RentMyItems): These platforms allow individuals to rent items from one another, but they are often limited to niche markets (e.g., tools, cameras) and do not cater specifically to the need for household item rentals. Furthermore, such platforms often lack user verification processes, which can reduce trust and make users hesitant to participate.

- Strengths: Peer-to-peer model, often lower costs than businesses.
- Weaknesses: Limited item categories, lack of trust mechanisms (e.g., verification of item condition), limited geographic availability.

3.2.4 Key Limitations of Existing Systems

1. Lack of Specialized Focus: Current platforms do not cater specifically to short-term residents who need household item rentals, leaving a critical gap for those in temporary living situations, such as NYSC participants and other short-stay occupants.
2. Limited Regional Coverage: Many rental services are unavailable in certain regions or fail to offer an adequate variety of essential household items, making it challenging for users to meet their specific needs.
3. Insufficient Trust Mechanisms: Existing peer-to-peer rental platforms often lack robust systems to verify the condition and quality of items, as well as the reliability of users, creating trust issues that deter participation.

These limitations highlight the need for a dedicated, secure, and user-friendly platform designed specifically to facilitate the short-term rental of household items, addressing both the logistical and trust challenges faced by users.

3.2.5 Requirements Gathering

This section outlines the essential requirements for the proposed system, both functional and non-functional, which will serve as the foundation for the system's design and implementation.

3.2.5.1 Functional Requirements

Functional requirements specify what the system must do. They define the system's features and the specific tasks it should perform. Based on the survey and interviews, the following functional requirements were identified:

1. User Authentication: The system must support user registration and login functionalities. This includes:

- Account creation using email or phone number.
- Password recovery options.
- Secure authentication mechanisms to protect user data.

2. Item Listings and Search: Users should be able to list items for rent and search for items based on specific filters, such as:
   - Item category (e.g., kitchen appliances, furniture, electronics).
   - Item condition and availability.
   - Location and delivery options.
   - Budget
3. Rental Transactions: The system must facilitate rental agreements, including:
   - Item rental duration.
   - Payment processing through secure channels.
   - A contract or rental agreement that outlines the terms and conditions.
   - Option for users to set security deposits for rented items.
4. Item Verification: A verification process should ensure that listed items are in good working condition. This will involve:
   - A quality check procedure conducted before items are listed for rent.
   - An additional quality check carried out when a borrower signifies interest in an item.
   - A final quality check to evaluate the condition of items before return.

5. Ratings and Reviews: After each rental, users should be able to rate both the item and the rental experience. This feature will help build trust among users and ensure the quality of the service.
6. Notifications and Alerts (Email): The system should send notifications to users about:
   - Upcoming rentals, due dates, and payments.
   - New items available for rent that match the user’s preferences.

3.2.5.2 Non-Functional Requirements

Non-functional requirements define the system's operational qualities, ensuring it meets performance and user experience standards.

1. Security: The system must provide robust security features to protect sensitive user data, including:
2. Secure payment processing and protection of financial data.
3. Scalability: The system should be designed to scale as the user base grows, capable of handling increasing numbers of users, item listings, and transactions without compromising performance.
4. User-Friendliness: The platform should be easy to navigate with a simple, intuitive interface. It should be responsive, ensuring it works well on both mobile and desktop devices.
5. Performance: The system should have fast load times for pages and transactions. Delays in listing items, searching for items, or completing rentals can frustrate users and deter them from using the platform.
6. Reliability and Availability: The platform should be highly reliable with minimal downtime, ensuring that users can access it at any time.
7. Legal Compliance: The system should comply with legal standards, such as user data protection laws, rental agreements, and security deposit regulations.

3.2.6 Stakeholder Input

Stakeholder input, derived from surveys and interviews, played a critical role in shaping the system's requirements. The responses from the questionnaire respondents revealed several key insights:

100% of respondents frequently find it too costly to purchase items for short-term use.

66.7% of respondents indicated that it is challenging to find short-term rentals for household items.

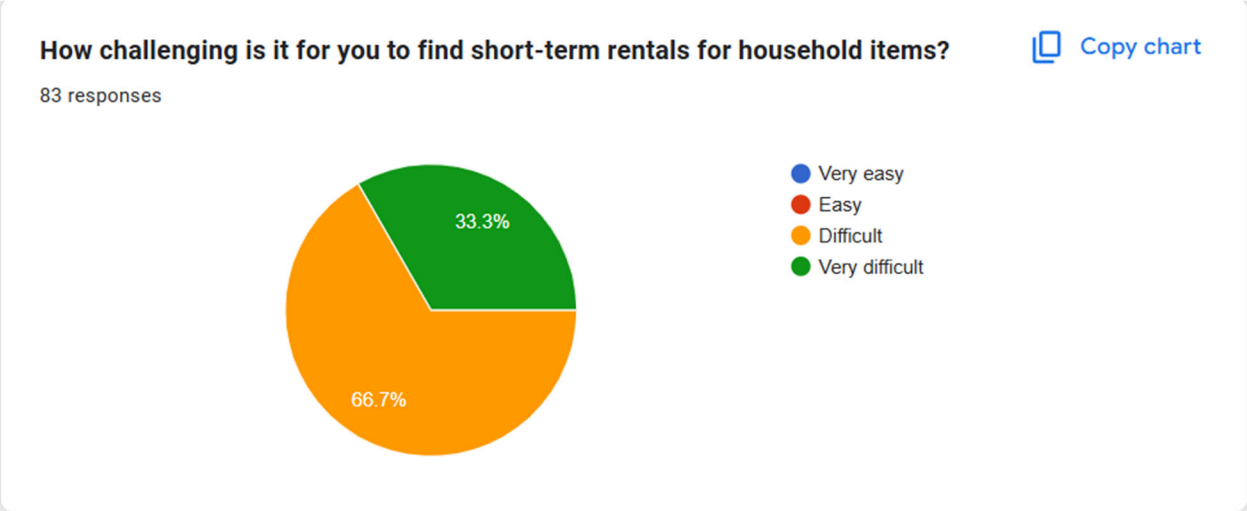


Figure 3.4: Survey statistic 4

Figure 3.4 shows the survey statistic for a particular question

66.7% of respondents stated they would feel slightly more comfortable renting from individuals if there was a verification process in place.

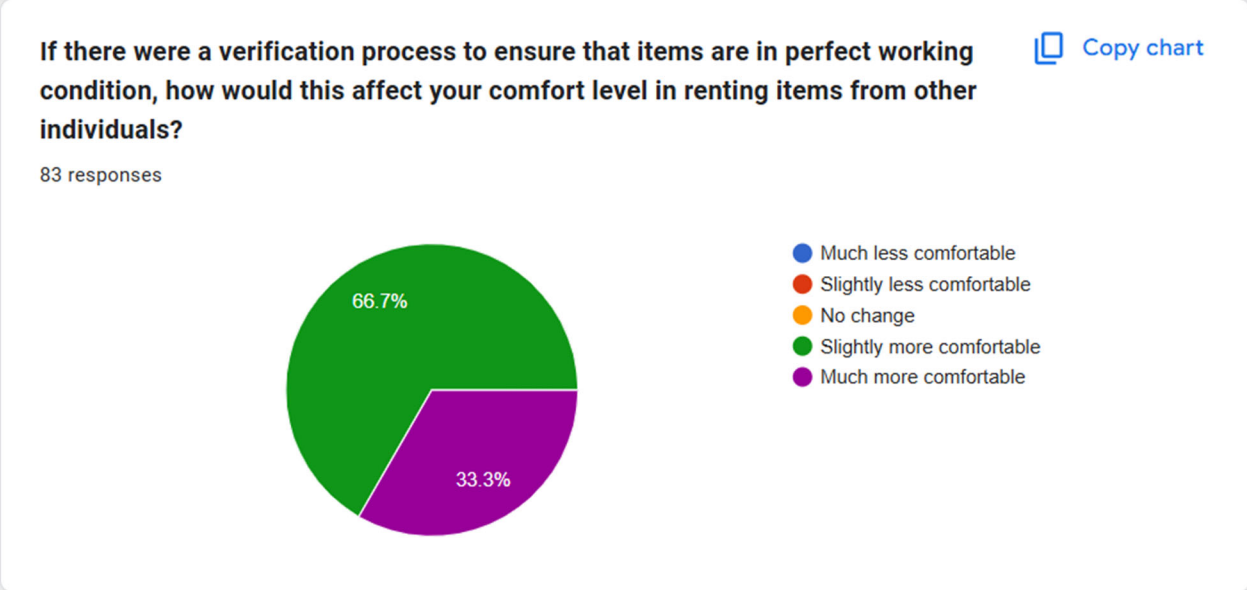


Figure 3.5: Survey statistic 5

Figure 3.5 shows the survey statistic for a particular question

In addition to the survey, interviews conducted with NYSC participants and other short-stay occupants provided deeper qualitative insights into the challenges they face and the expectations they have for a rental platform. Respondents consistently highlighted several pain points and suggested features that would make the proposed system effective:

- Verification Processes: A respondent mentioned that having a verification process to confirm the condition of items before rental would make them more comfortable using the platform. This was echoed by several others, who highlighted trust as a critical factor in their willingness to participate.
- User Authentication: One respondent suggested that ensuring users are verified through a secure process would increase confidence in renting from individuals rather than businesses.
- Secure Payment Methods: A participant noted that a seamless and secure payment method is essential, particularly to avoid disputes over rental payments or refunds.
- Security Features: Another respondent emphasized the importance of features like damage deposits or insurance to protect both renters and lenders, which would encourage more people to use the system.

These insights were instrumental in defining the system's features, such as the need for a verification process for items, user authentication, secure payment methods, and the inclusion of security features to protect both parties involved in transactions.

In conclusion, this section has provided a detailed analysis of the problem, evaluated the limitations of existing systems, and outlined the functional and non-functional requirements that will guide the development of the proposed platform. The findings from the surveys and interviews underscore the pressing need to address the challenges faced by short-term residents. These insights highlight the necessity of a peer-to-peer rental platform that delivers convenience, security, and affordability.

The next section will focus on the design of this system, building upon the requirements established here.

3.3 SYSTEM DESIGN:

The system design chapter provides a detailed explanation of the architecture, components, and workflows of the proposed web-based peer-to-peer (P2P) rental platform for household items. This chapter is structured into several sections, each addressing specific aspects of the system's design, including use-case diagrams, system architecture, activity diagrams, sequence diagrams, entity-relationship models, data models, and class diagrams. The goal is to provide a comprehensive blueprint for the development of the platform, ensuring that it meets the functional and non-functional requirements outlined in the previous chapters.

3.3.1 Use-Case Diagram

The use-case diagram provides a high-level overview of the system's interactions, illustrating how users (actors) interact with the platform's core functionalities. The diagram identifies two primary actors: the Owner (lender) and the Renter (borrower). These actors interact with the system through various use cases, which represent the platform's key functionalities.

The primary use cases include:

- Item Listing: Owners can list household items for rent by providing details such as item category, price, availability, and images.
- Item Search: Renters can search for available items based on filters like location, category, and price range.
- Rental Request: Renters can request to rent an item, specifying the rental duration and agreeing to the terms.
- Payment Processing: The system facilitates secure payments between renters and owners using a payment gateway.
- Rating and Reviews: After a rental transaction, both renters and owners can rate each other and provide feedback.

The relationships between the actors and use cases illustrate the possible actions users can perform within the system. For example, an owner can list an item, while a renter can search for and request to rent that item. This diagram serves as a foundation for understanding the system's functionality from a user perspective.

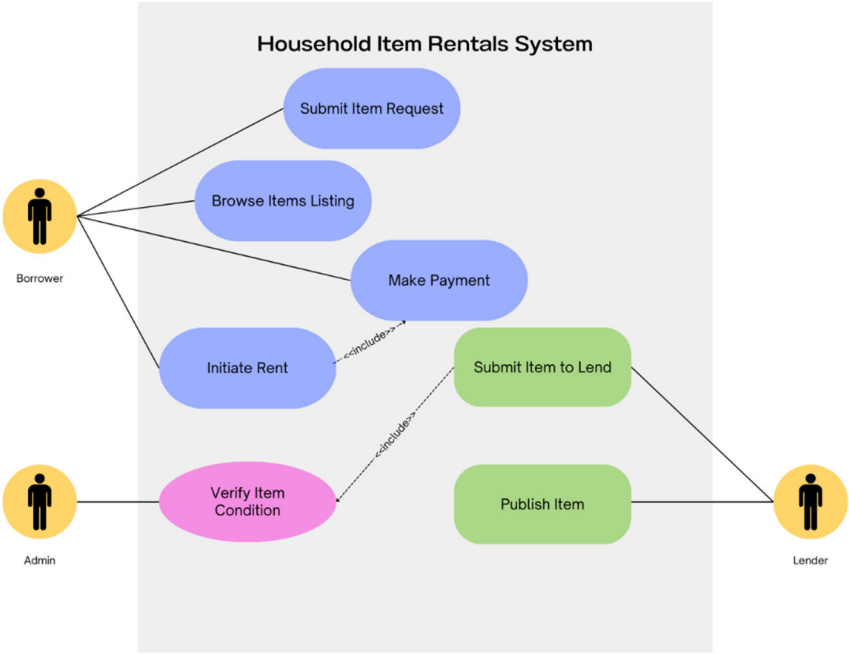


Figure 3.6: Use-Case diagram

Figure 3.6 shows the use-case diagram of the system

3.3.2 System Architecture

The system architecture outlines the major components of the platform, their interactions, and how data flows within the system. The architecture is divided into several key components:

1. User Interface:

   The user interface (UI) is the primary point of interaction between users and the system. It provides an intuitive and responsive interface for users to perform actions such as browsing items, listing items, and managing rentals. The UI ensures a seamless user experience, with features like search functionality, item details, and rental request forms.

2. Application Logic:

   The application logic handles the core functionalities of the platform, including user authentication, item management, rental requests, and payment processing. It ensures that all user actions are processed correctly and that data is validated before being stored or transmitted.

3. Database:

   The database stores all critical data, including user information, item details, rental transactions, and reviews. It is designed to support efficient data retrieval and storage, ensuring that the platform can handle a large number of users and transactions.

4. Payment Gateway:

   The payment gateway facilitates secure transactions between renters and owners. It ensures that payments are processed securely and that financial data is protected. The gateway also handles refunds and security deposits, providing a reliable payment solution for users.

5. Storage:

   The storage component is responsible for managing images and other essential assets uploaded by users. It ensures that item images, user profile pictures, and verification documents are stored securely and can be accessed efficiently.

The system architecture ensures that all components work together seamlessly to provide a reliable and efficient platform for household item rentals.

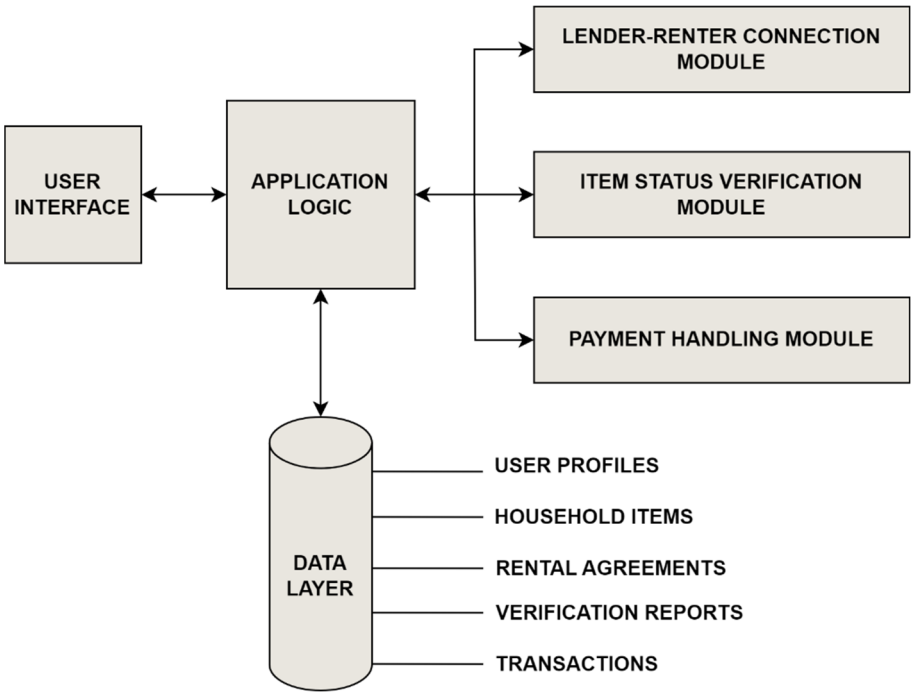


Figure 3.7: High level System Architecture

Figure 3.7 outlines the high-level system architecture

3.3.3 User Interface

The user interface (UI) is the point of interaction between users and the system. A low-fidelity paper prototype is used to design the core pages of the application. The UI design focuses on simplicity, usability, and responsiveness. The low-fidelity paper prototype includes the following core pages:

1. Homepage (Landing Page):

   The homepage serves as the entry point for users. It provides an overview of the platform's features and includes:

   - A search bar for finding items.
   - Featured items or categories.
   - Links to login/signup pages.

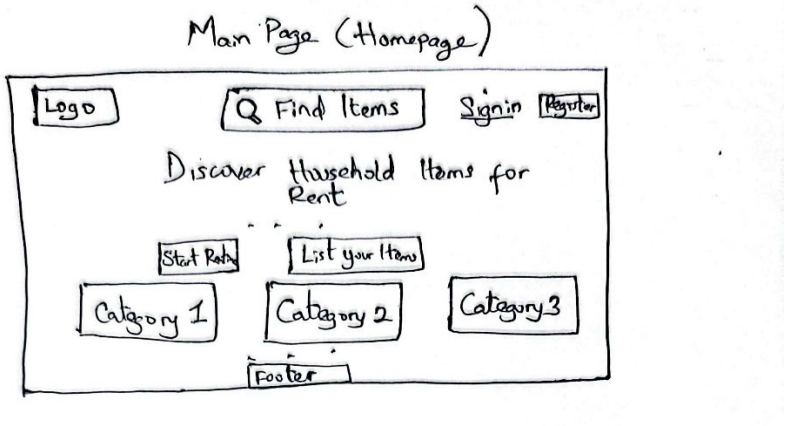


Figure 3.8: UI - Homepage (Paper Prototype)

Figure 3.8 presents the paper prototype of the homepage

2. Item List Page:

   This page displays a list of available items based on the user's search criteria. Key features include:

- Filters for category, price range, and location.
- Item cards with basic details (image, name, price, location).
- Pagination or infinite scrolling for browsing items.

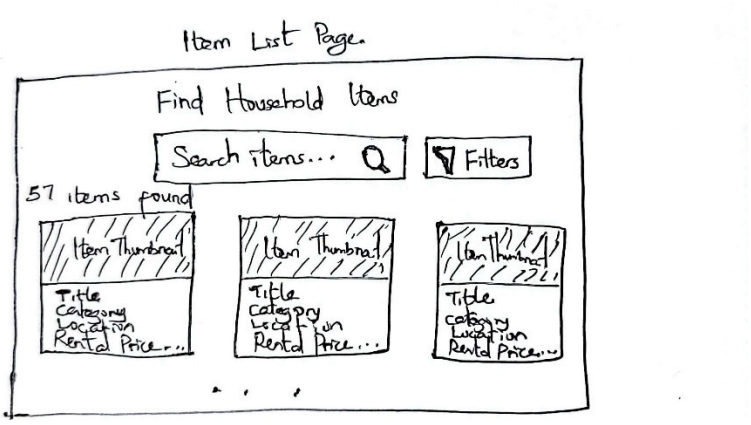


Figure 3.9: UI - Item list page (Paper Prototype)

Figure 3.9 displays the paper prototype of the item list page

3. Item Detail Page:

   This page provides detailed information about a specific item. It includes:

   - Item images and description.
   - Rental terms (price, duration, availability).
   - Owner details and ratings.
   - A "Book Item Now" button.

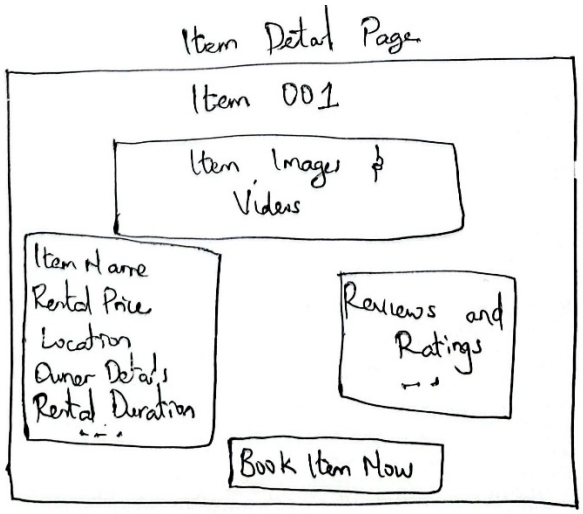


Figure 3.10: UI - Item detail page (Paper Prototype)

Figure 3.10 illustrates the paper prototype of the item detail page

4. Item Upload Form:

   This form allows owners to list new items for rent. It includes fields for:

   - Item name, category, and description.
   - Rental price and availability.
   - Uploading images.
   - Setting rental terms (e.g., security deposit, minimum rental duration).

The low-fidelity prototype ensures that the core functionalities are well-defined before moving to high-fidelity designs and development.

3.3.4 Activity Diagrams

3.3.4.1 Activity Diagram - Item Request

This diagram represents the sequence of steps involved when a renter requests an item. The process begins with the renter browsing available items on the platform. Once the renter selects an item, the system checks if the item is available and if the renter agrees to the rental terms. If both conditions are met, the owner is notified of the request. The owner can then approve or decline the request. If approved, the renter proceeds to make payment, and the rental agreement is finalized. If declined, the renter is notified, and the process ends.

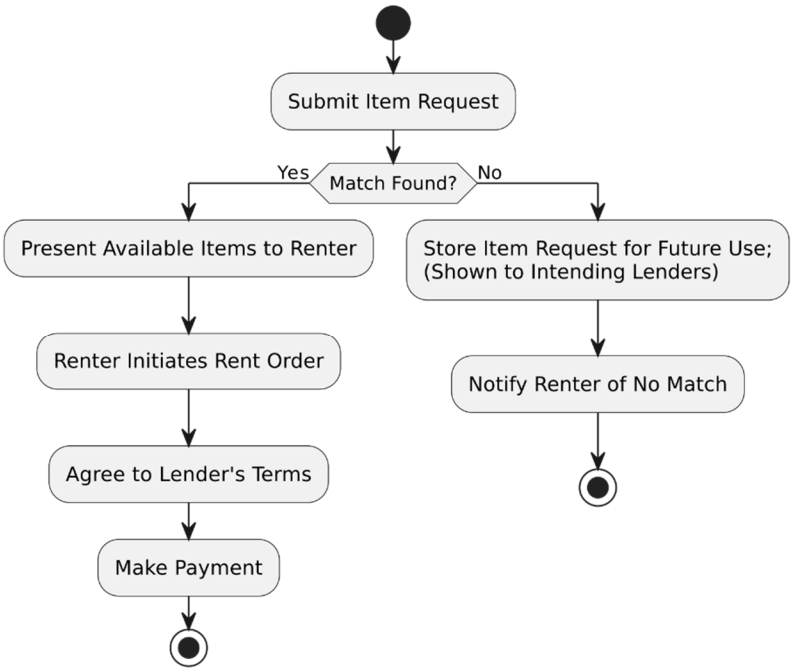


Figure 3.11: Activity diagram - Item Request

Figure 3.11 shows the activity diagram for item requests

3.3.4.2 Activity Diagram - Item Upload

This diagram shows the process of an owner listing an item for rent. The process starts when the owner selects the “Add Item” function. The owner then enters item details, such as category, price, availability, and images. The system checks if all required fields are completed before allowing the owner to submit the listing. Once submitted, the item is stored in the database and becomes searchable by renters.

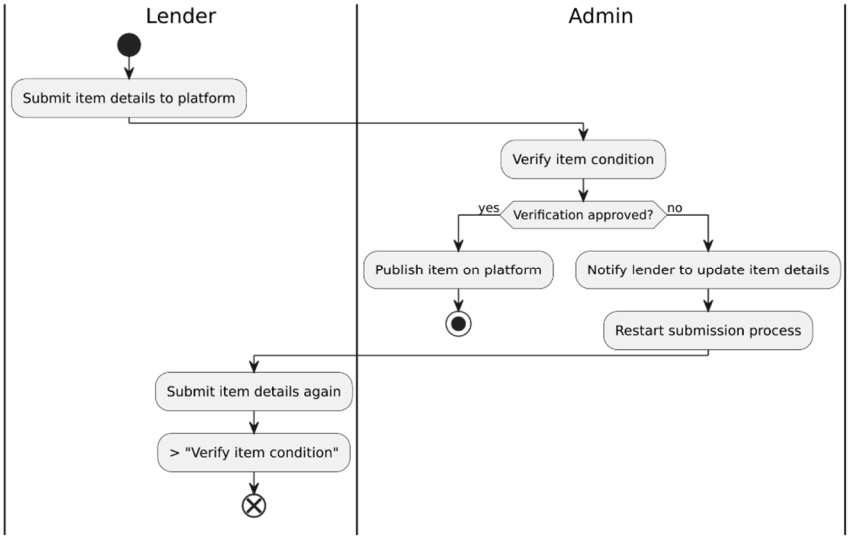


Figure 3.12: Activity diagram - Item Upload

Figure 3.12 presents the activity diagram for item uploads

3.3.5 Sequence Diagrams

3.3.5.1 Sequence Diagram - Item Request and Matching

This diagram illustrates how objects interact during the rental request process. The sequence begins when the renter sends a rental request for an item. The system processes the request and notifies the owner. The owner reviews the request and responds with either approval or rejection. The system then updates the rental status and notifies the renter of the outcome.

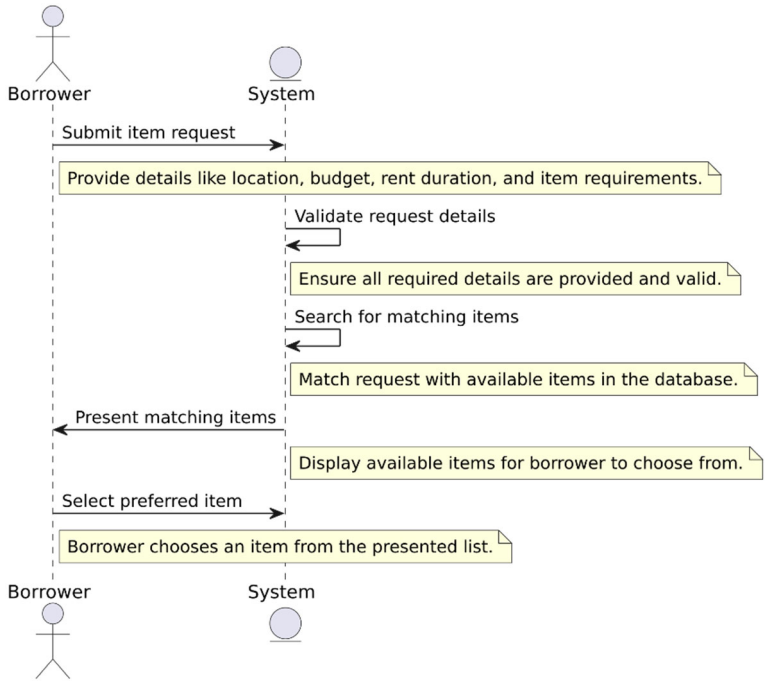


Figure 3.13: Sequence diagram - Item Request and Matching

Figure 3.13 illustrates the sequence diagram for item request and matching

3.3.5.2 Sequence Diagram - Item Upload

This diagram depicts how the owner interacts with the system to upload an item. The owner inputs item details, including images and rental terms. The system validates the input to ensure all required fields are completed. If validation is successful, the item is stored in the database and made visible to renters.

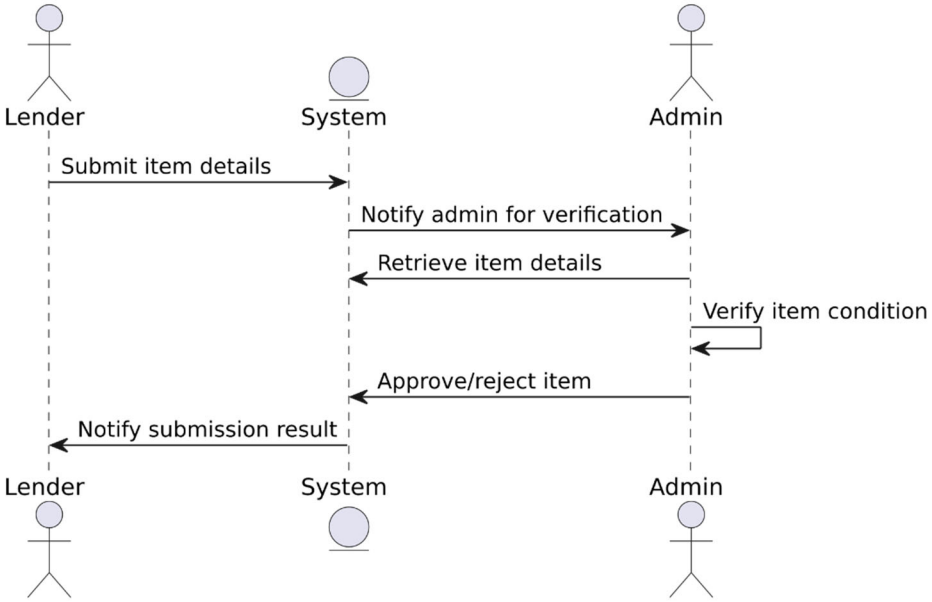


Figure 3.14: Sequence diagram - Item Upload

Figure 3.14 shows the sequence diagram for item uploads

3.3.5.3 Sequence Diagram - Rent Initiation

This diagram illustrates the process of finalizing a rental agreement after an item request is approved. Once the owner approves a rental request, the renter makes payment through the platform. The system verifies the payment and notifies the owner. A rental contract is generated, and the transaction is confirmed.

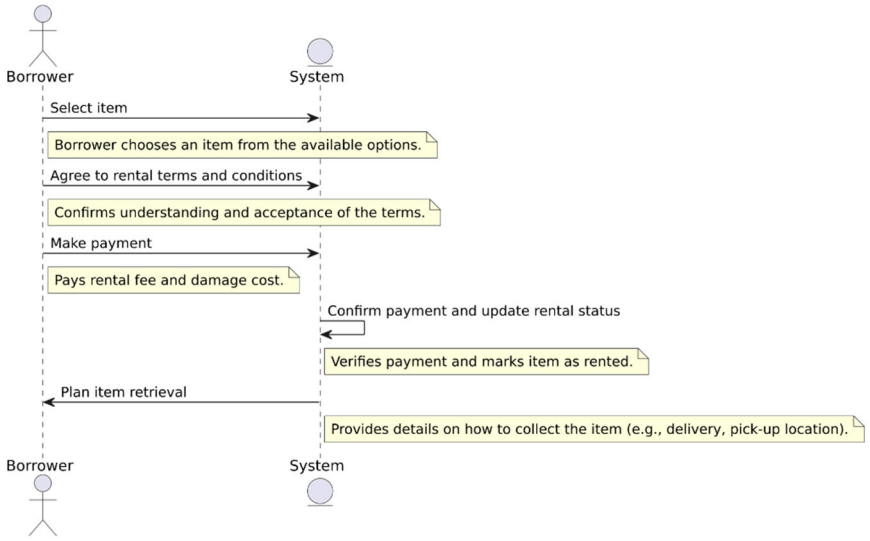


Figure 3.15: Sequence diagram - Rent Initiation

Figure 3.15 depicts the sequence diagram for rent initiation

3.3.5.4 Sequence Diagram - Item Verification on Rent Order

This diagram shows the process of verifying an item's condition before it is handed over to the renter. Before the rental begins, admin inspects the item to ensure it is in good condition. The system records the item's condition, and the renter receives the item along with a condition report.

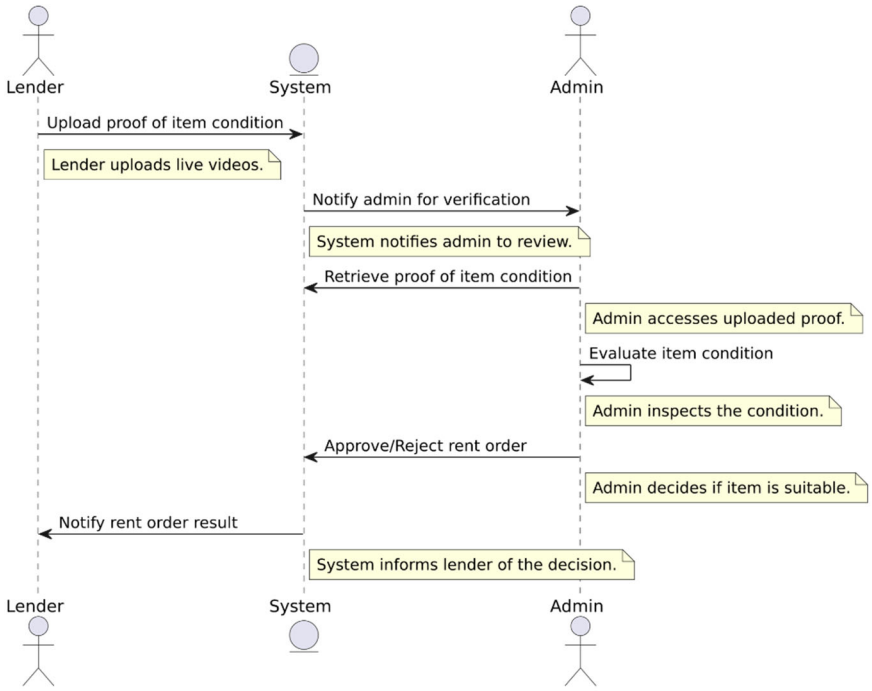


Figure 3.16: Sequence diagram - Item Verification on Rent Order

Figure 3.16 illustrates the sequence diagram for item verification on rent order

3.3.5.5 Sequence Diagram - Item Verification Before Return

This diagram describes how the system ensures the item is returned in proper condition. At the end of the rental period, admin inspects the item against the rental terms to verify its condition. If the item is in good condition, the security deposit is refunded to the renter. If issues arise, the system initiates a dispute resolution process.

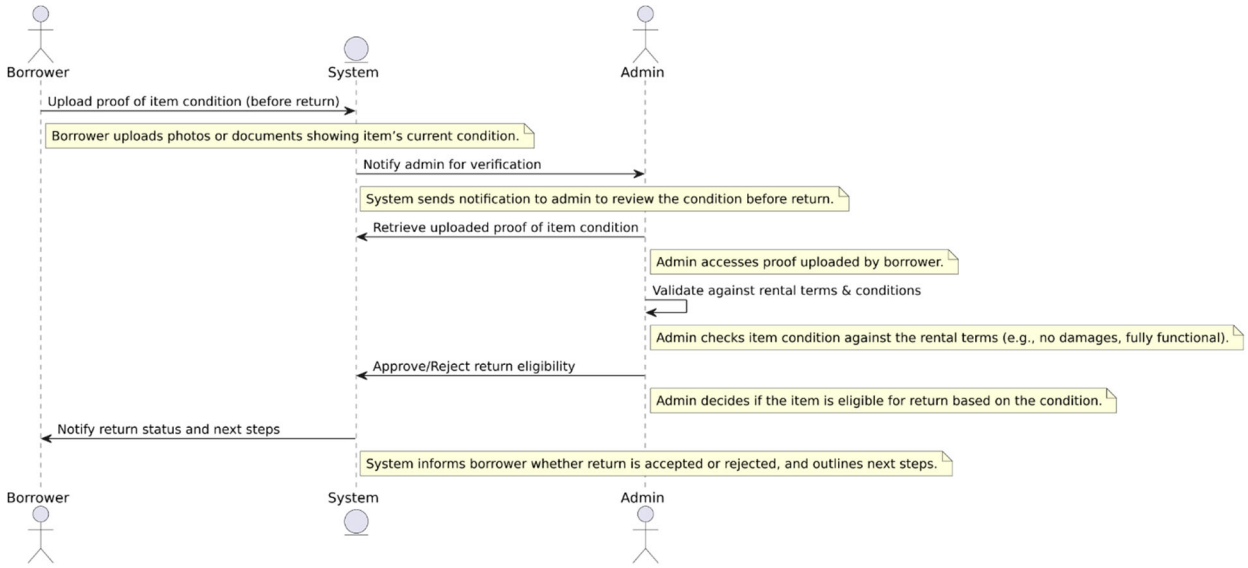


Figure 3.17: Sequence diagram - Item Verification before Return

Figure 3.17 shows the sequence diagram for item verification before return

3.3.6 Data Flow Diagram

3.3.6.1 Context Diagram (Level 0)

The Context Diagram (Level 0) offers a high-level overview of the Household Item Rental System, showing interactions with three external entities—**User** (Owner, Renter, Admin), **Payment Gateway**, and **Scheduling Tool**. The User sends registration, listing, rental, payment, and review data to the system, receiving confirmations, search results, and meeting links in return. The **Payment Gateway** processes payment authorizations and refunds, returning confirmations and statuses. The **Scheduling Tool** receives time slots and attendee details, providing scheduled meeting links. All flows converge on the central system, which manages these processes without revealing internal logic at this level.

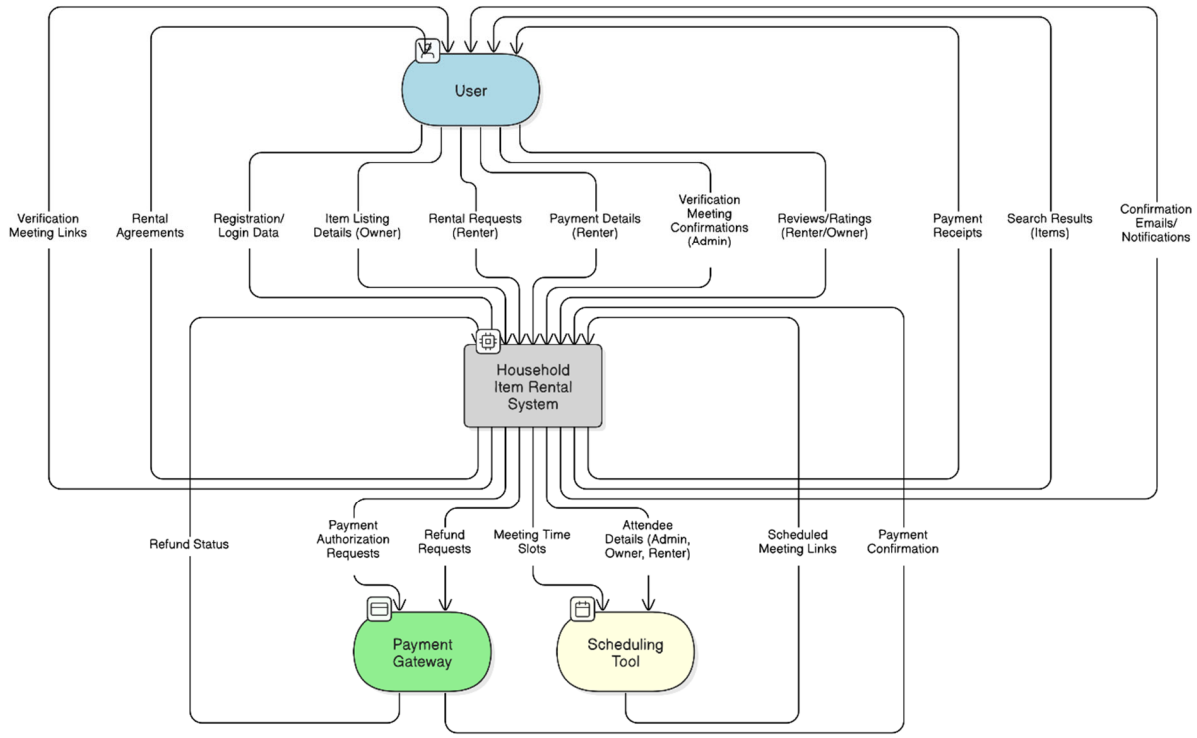


Figure 3.18: Context Diagram (Level 0 DFD)

Figure 3.18 presents the context diagram of the system

3.3.6.2 Level 1 DFD

At this level, the Household Item Rental System is broken down into sub-processes for user registration, verification meeting management, payment processing, item listing, item searching, rental requests, and review submission. Each sub-process exchanges data with the User (in any role), as well as with external services like the Payment Gateway and the Scheduling Tool, ensuring that registration/login data, payment details, meeting schedules, and reviews flow appropriately between the system and its data stores.

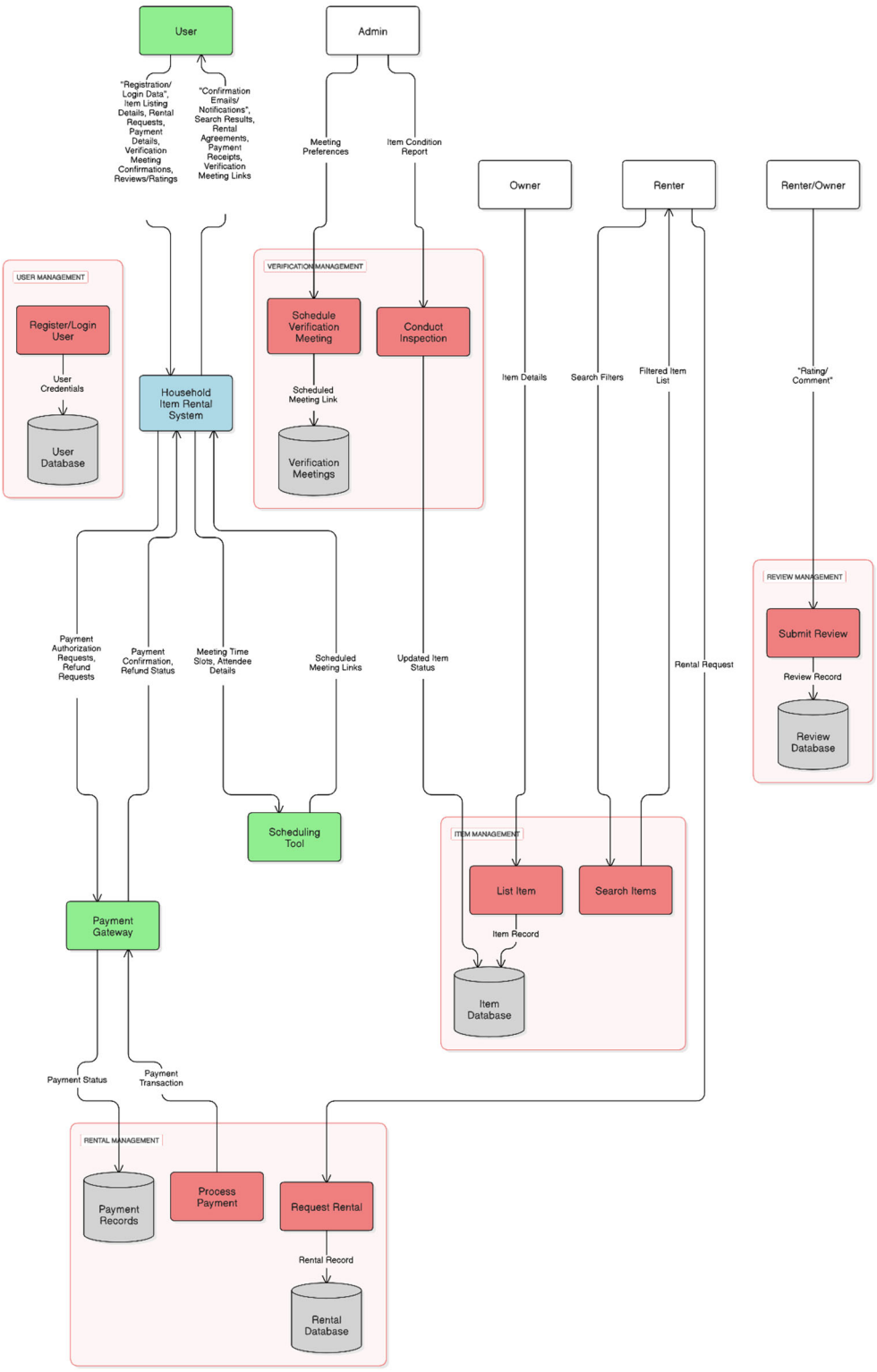


Figure 3.19: Level 1 Data Flow Diagram

Figure 3.19 displays the level 1 data flow diagram of the system

3.3.7 Entity-Relationship (ER) Model

The ER model defines the database structure and illustrates how the entities interact through defined relationships and cardinalities. Each entity is represented as a table with its attributes, and the relationships specify how many instances of one entity can be associated with instances of another. The key entities and their relationships are described below:

- **Users Table**:

  This table stores information for all system users. It includes attributes such as userID, email, passwordHash, fullName, phoneNumber, role, createdAt, and updatedAt.

  Relationship & Cardinality:

  Each user is specialized into one of three subtypes — Owner, Renter, or Admin — represented by a one-to-one (1:1) relationship:

  - USER ||--|| OWNER : "is a"
  - USER ||--|| RENTER : "is a"
  - USER ||--|| ADMIN : "is a"

- **Owner Table**:

  Contains additional information for users who list items for rent, such as bankAccountNumber and rating.

  Relationship & Cardinality:

  Each Owner is linked to exactly one User. An Owner can list multiple Items, representing a one-to-many (1:Many) relationship:

  - OWNER ||--o{ ITEM : "lists"

- **Renter Table**:

  Stores details for users who rent items, including their address and verificationStatus.

  Relationship & Cardinality:

  Each Renter is linked to exactly one User, and a Renter can request multiple Rentals (1:Many):

  - RENTER ||--o{ RENTAL : "requests"

- **Admin Table**:

  Contains information about administrators, such as privilegeLevel.

  Relationship & Cardinality:

Each Admin is linked to exactly one User and can manage multiple Verification Meetings (1:Many):

- USER ||--|| ADMIN : "is a"
- ADMIN ||--o{ VERIFICATION_MEETING : "manages"

- **Item Table**:

  Represents rentable items with attributes like name, description, rentalTerms, dailyPrice, availabilityStatus, condition, createdAt, and updatedAt.

  Relationship & Cardinality:

  Each Item is associated with one Owner (an Owner can list many Items), and each Item belongs to one Category.

  - OWNER ||--o{ ITEM : "lists"
  - ITEM ||--|{ CATEGORY : "belongs to"

- **Rental Table**:

  Captures rental transactions, linking a Renter to an Item along with rental details such as startDate, endDate, totalCost, status, and createdAt.

  Relationship & Cardinality:

  Each Rental is linked to one Renter and one Item, but an Item can be rented multiple times (1:Many). Additionally, each Rental can have associated Payments, Verification Meetings, Reviews, and a Damage Deposit.

  - RENTER ||--o{ RENTAL : "requests"
  - ITEM ||--o{ RENTAL : "is rented in"
  - RENTAL ||--|{ PAYMENT : "has"
  - RENTAL ||--o{ VERIFICATION_MEETING : "requires"
  - RENTAL ||--|{ REVIEW : "generates"
  - RENTAL ||--|{ DAMAGE_DEPOSIT : "holds"

- **Payment Table**:

  Records payments linked to Rentals, with attributes such as amount, paymentMethod, transactionReference, status, and timestamp.

  Relationship & Cardinality:

  A Rental can have multiple Payments (1:Many).

- **Verification Meeting Table**:

Represents meetings conducted for verifying item conditions (pre-rental, on-rent, and post-rental), including details like type, scheduledTime, status, notes, and calendarLink.

Relationship & Cardinality:

Each Verification Meeting is linked to one Rental, and an Admin can manage multiple meetings (1:Many).

- **Review Table**:

  Contains feedback on Rentals, with attributes such as rating, comment, and createdAt.

  Relationship & Cardinality:

  Each Rental generates exactly one Review (1:1).

- **Category Table**:

  Classifies items (e.g., Furniture, Electronics, Kitchen Appliances) with attributes like name and description.

  Relationship & Cardinality:

  Each Item belongs to one Category, while a Category can include multiple Items (1:Many).

- **Damage Deposit Table**:

  Stores information on security deposits for Rentals, including amount, refundStatus, and refundDate.

  Relationship & Cardinality:

  Each Rental holds exactly one Damage Deposit (1:1).

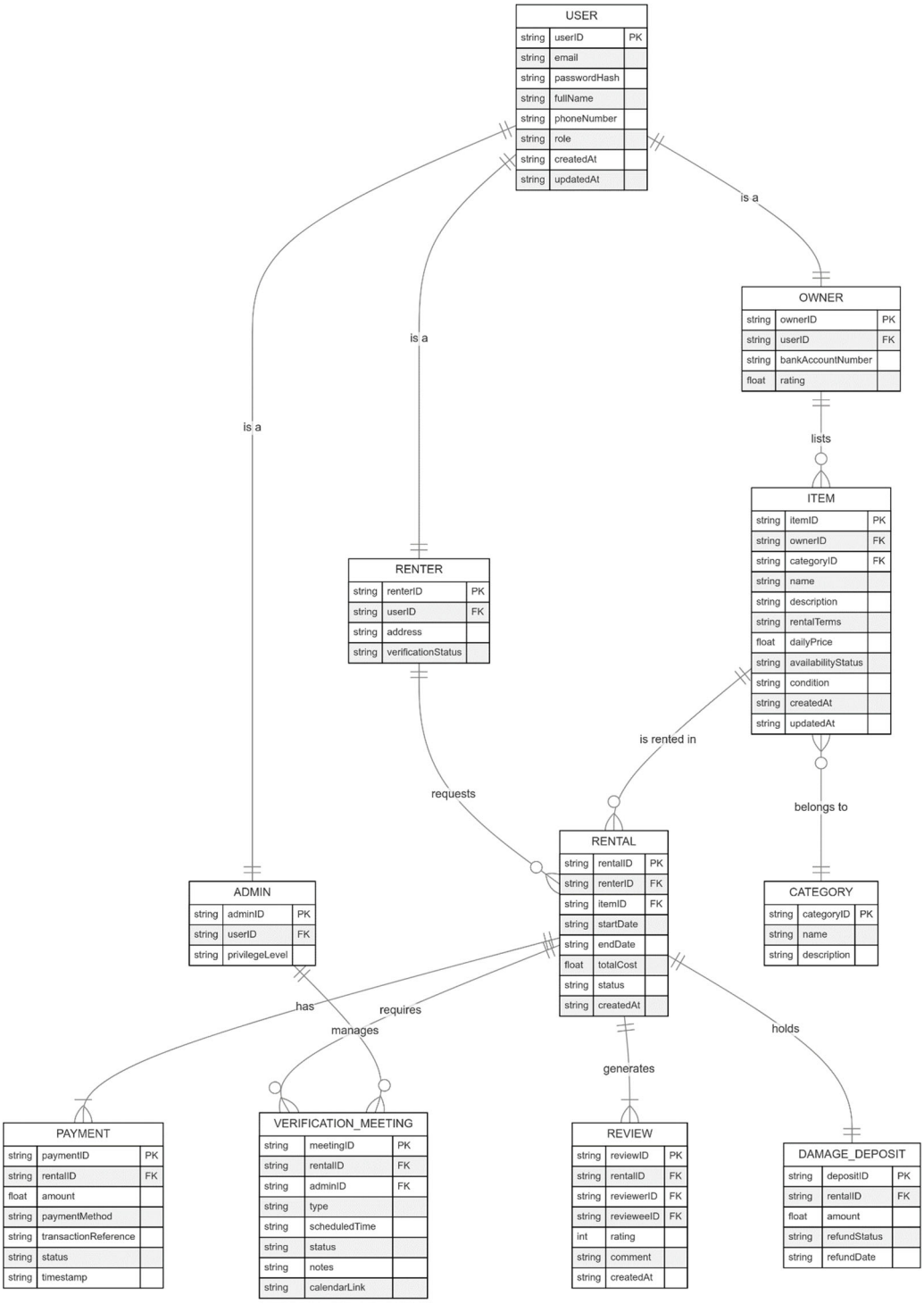


Figure 3.20: Entity-Relationship diagram

Figure 3.20 illustrates the entity-relationship model of the system

3.3.8 Class Diagram

The class diagram provides an object-oriented view of the HiRentals Household Items Rental system, outlining the main classes, their attributes, behaviors (functions), and relationships. The diagram models core entities such as User, Item, Rental, Payment, VerificationMeeting, Review, Category, and DamageDeposit.

- **User and Specializations**:

  The User class encapsulates common attributes like userID, email, and contact details. This class is specialized into three distinct roles—Owner, Renter, and Admin—using inheritance, ensuring that each user is uniquely classified. Each specialization adds role-specific attributes and methods (e.g., an Owner can list items, a Renter can request rentals, and an Admin manages verification meetings).
- **Item and Rental Management**:

  The Item class represents rentable items, including details such as name, description, rental terms, and status. The Rental class links a Renter to an Item and captures transaction details such as rental duration and cost. Functions within these classes handle actions like starting or ending rentals and updating item availability.
- **Transaction and Verification**:

  The Payment class models payment transactions related to rentals, incorporating functions to process and refund payments. VerificationMeeting manages pre- and post-rental inspections, with methods for scheduling meetings and recording outcomes, ensuring that items are properly verified.
- **Feedback and Security**:

  The Review class allows renters to rate and comment on their rental experience, while the DamageDeposit class deals with security deposits, enforcing deposit holds and refunds based on the condition of the rented items.
- **Categorization**:

  The Category class organizes items into groups (e.g., Furniture, Electronics) and provides functionality for assigning and updating category details.

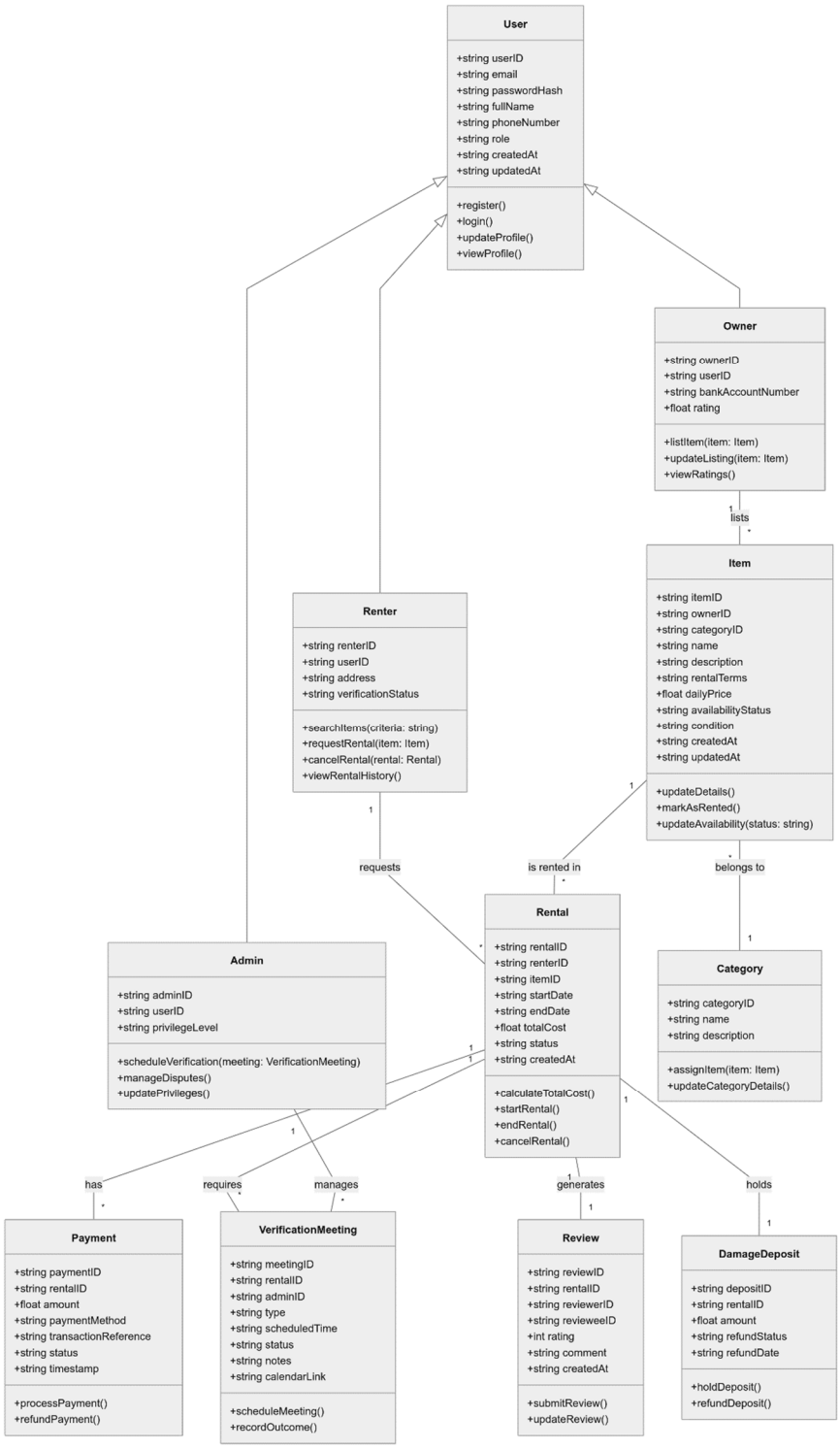


Figure 3.21: Class diagram

Figure 3.21 shows the class diagram of the system

3.4 SUMMARY:

This chapter has provided a comprehensive overview of the System Analysis and System Design for the proposed web-based peer-to-peer (P2P) rental platform for household items. The methodology adopted in this project ensures that the system is developed systematically, addressing both user needs and technical requirements.

System Analysis focused on understanding the problem domain and gathering requirements from stakeholders. Through surveys and interviews, key insights were gathered, highlighting the financial and logistical challenges faced by short-term residents, such as students and temporary workers. The analysis identified the need for a platform that allows users to rent household items on a temporary basis, without the burden of ownership or expensive logistics. Functional and non-functional requirements were defined, ensuring that the system meets user expectations in terms of usability, security, and scalability.

System Design provided a high-level blueprint for the platform, outlining its architecture, components, and workflows. The design phase included the creation of use-case diagrams, activity diagrams, sequence diagrams, entity-relationship models, data models, and class diagrams. These diagrams collectively define how users interact with the system, how data flows between components, and how the system ensures reliability and security. Key components of the system include the User Interface, Application Logic, Database, Payment Gateway, and Storage. The design also incorporates trust-building mechanisms, such as item verification and secure payment processing, to ensure a seamless and trustworthy user experience.

In summary, the System Analysis phase laid the groundwork for understanding user needs and defining system requirements, while the System Design phase translated these requirements into a detailed architectural plan. Together, these phases ensure that the proposed platform is both functional and user-centric, addressing the challenges faced by short-term residents in accessing household items. The next chapter will focus on the Implementation and Testing of the system, bringing the design to life and validating its functionality.

CHAPTER FOUR:
IMPLEMENTATION AND TESTING

4.1 INTRODUCTION:

This chapter provides a detailed account of the implementation and testing phases of the **Household Items Rental** system, a web-based solution designed to facilitate the rental of household items between users. The application is built to provide an intuitive and seamless experience for users and is fully functional, with a robust backend powered by Firestore for database storage and Firebase Storage for file management. It features automated meeting scheduling via Calendly, secure payment processing through Paystack, and comprehensive dashboards for Owners, Renters, and Admins. This chapter provides an in-depth analysis of the system's architecture, functionality, and testing strategies.

Terminologies Used:

- HiRentals: The application name.
- Owner: A user who lists and lends household items for rent.
- Renter: A user who borrows and rents household items for temporary use.

4.2 IMPLEMENTATION:

The HiRentals application is developed using TypeScript as the primary programming language, with additional support from JavaScript. The front-end is built using React.js, a popular JavaScript library for building user interfaces, while Firebase serves as the backend for data storage, authentication, and file storage.

The application is hosted on Vercel and is publicly accessible at: https://hirentals.vercel.app

The codebase is hosted on GitHub and can be accessed at: https://github.com/p-Smart/hirentals



Figure 4.1: QR Code for Live Application

Figure 4.1 presents the QR code for the live application



Figure 4.2: QR Code for Application's Source Code

Figure 4.1 presents the QR code for the application's source code

4.2.1 Technology Stack

The system was implemented using the following technologies:

- Frontend: React.js with TypeScript (UI components, state management)

- Backend: Firebase (Firestore for database, Firebase Functions for backend logic)
- Storage: Firebase Storage for handling item images and videos
- Authentication: Firebase Authentication (email/password, social login)
- Payment Processing: Paystack (secure transactions, automated refunds)
- Meeting Scheduling: Calendly (automatic appointment scheduling)
- Hosting & Deployment: Vercel (continuous deployment, performance optimization)
- Version Control & Code Management: GitHub (repository management, collaboration, version control)

4.2.2 Home Page

The Home Page serves as the entry point to the application, providing users with an overview of the services offered. It features a navigation bar for easy access to other sections of the application, a hero section with a call-to-action button, and a preview of available items. The design is clean and user-friendly, ensuring that users can quickly understand the purpose of the platform and navigate to the desired sections.

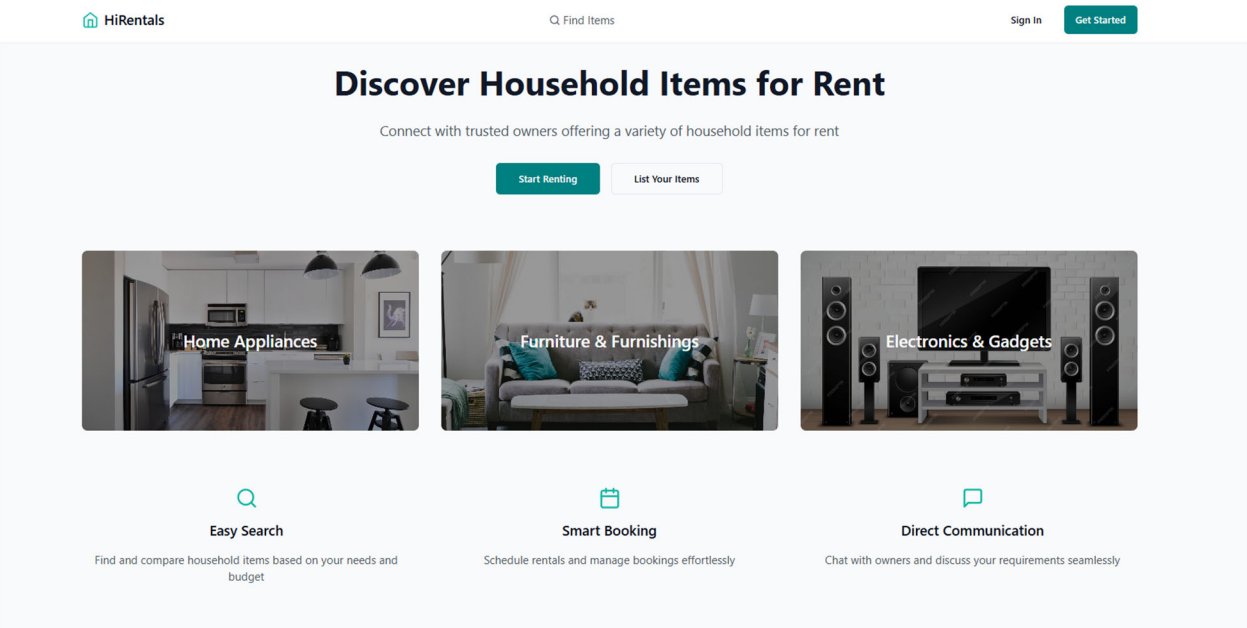


Figure 4.3: Homepage

Figure 4.3 shows the homepage layout

4.2.3 Item Listing Page

The Item Listing Page displays a comprehensive list of items available for rent. Users can filter items based on categories, price range, and availability. Each item is presented with an image, title, brief description, and rental price. The page is designed to be responsive, ensuring a seamless experience across devices, including desktops, tablets, and mobile phones.

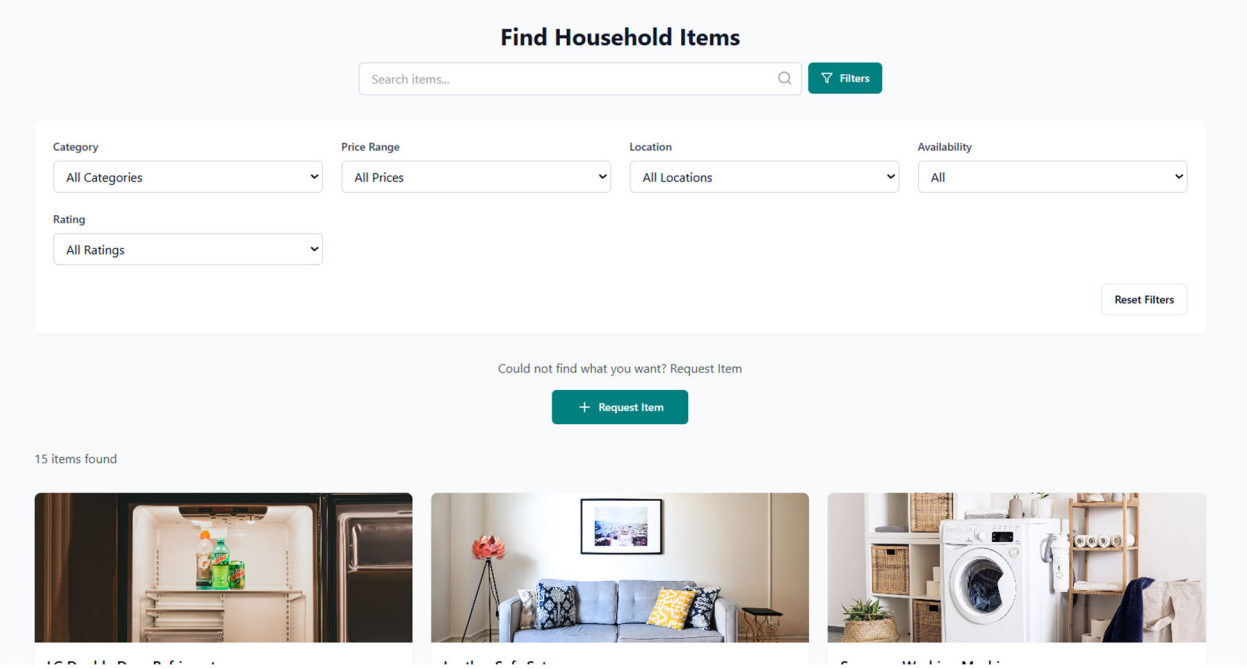


Figure 4.4: Item listing page 1

Figure 4.4 displays the first view of the item listing page

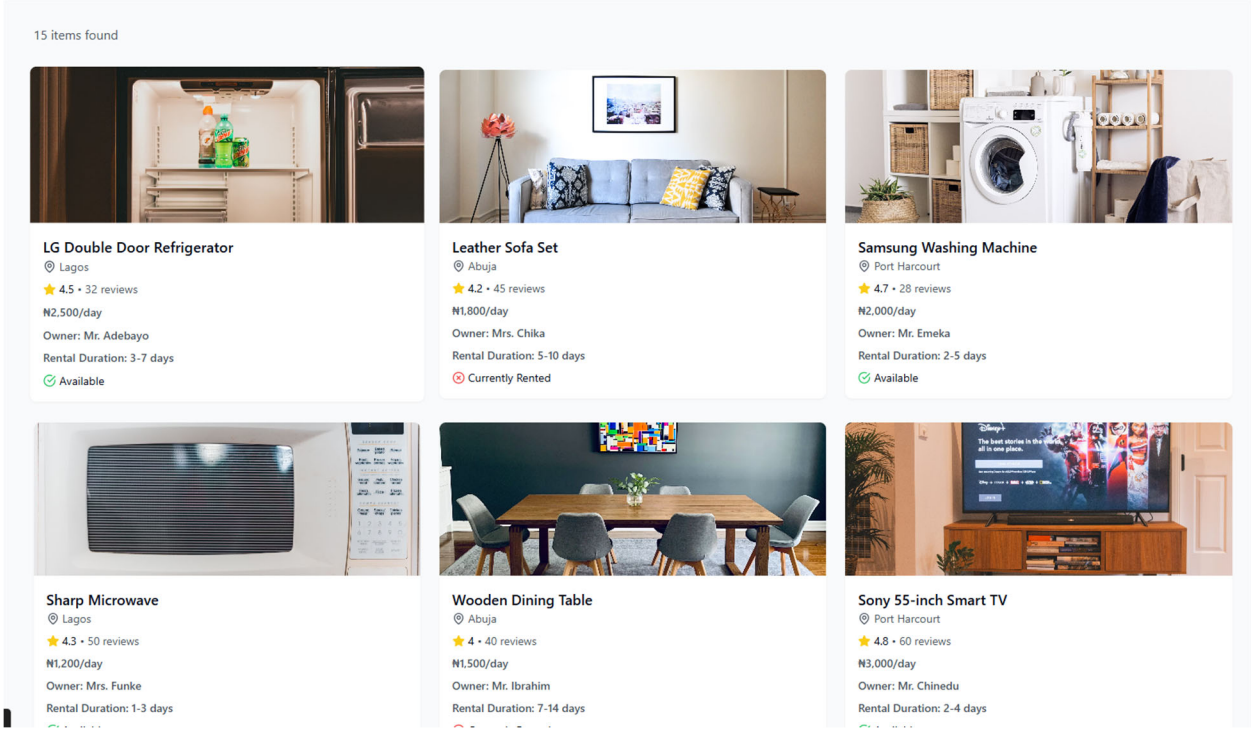


Figure 4.5 Item listing page 2

Figure 4.5 shows the second view of the item listing page

4.2.4 Item Request Form

The Item Request Form allows renters to request specific household items if they cannot find what they are looking for on the platform. This feature enhances user engagement and ensures a wider variety of items are available over time.

Functionality:

- Renters can fill out a request form specifying the item name, category, description, and preferred rental duration.
- Submitted requests are displayed on a dedicated "Requested Items" page within the Owners' Dashboard.
- Owners can view requested items and list an item for rent if they have it.

This feature helps bridge the gap between demand and supply, ensuring users get the household items they need while encouraging Owners to upload relevant inventory.

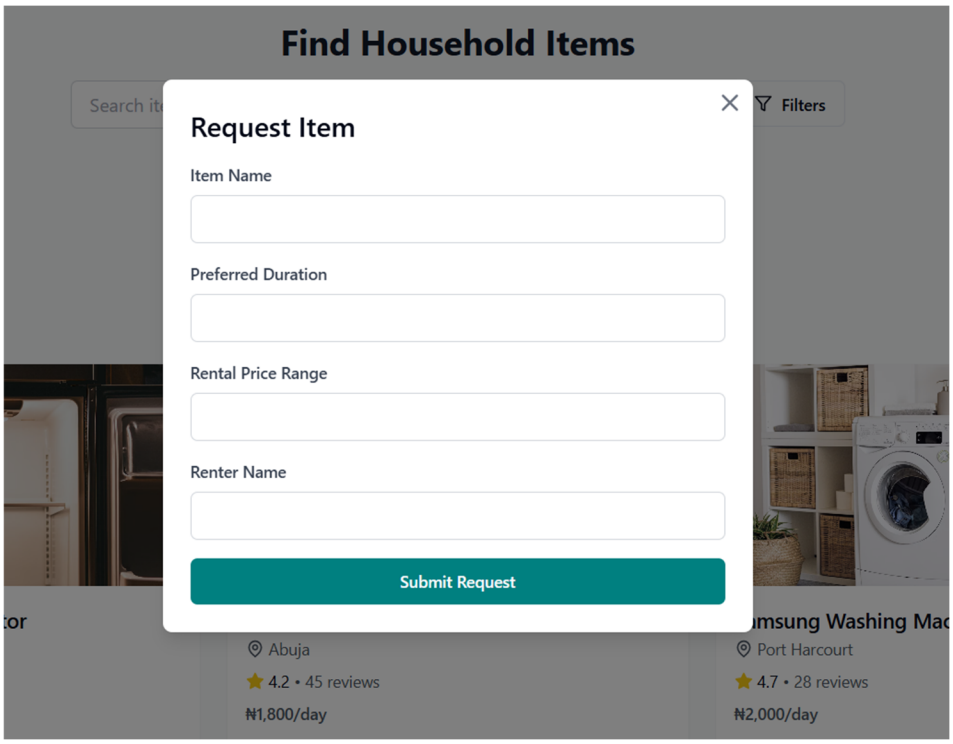


Figure 4.6: Item Request Form

Figure 4.6 shows the item request form

4.2.5 Item Details Page

The Item Details Page provides detailed information about a selected item. It includes multiple images, a detailed description, rental terms, and a booking button. Users can also view related items on this page, encouraging them to explore more options. The page is designed to provide all the necessary information for users to make informed decisions about renting an item.

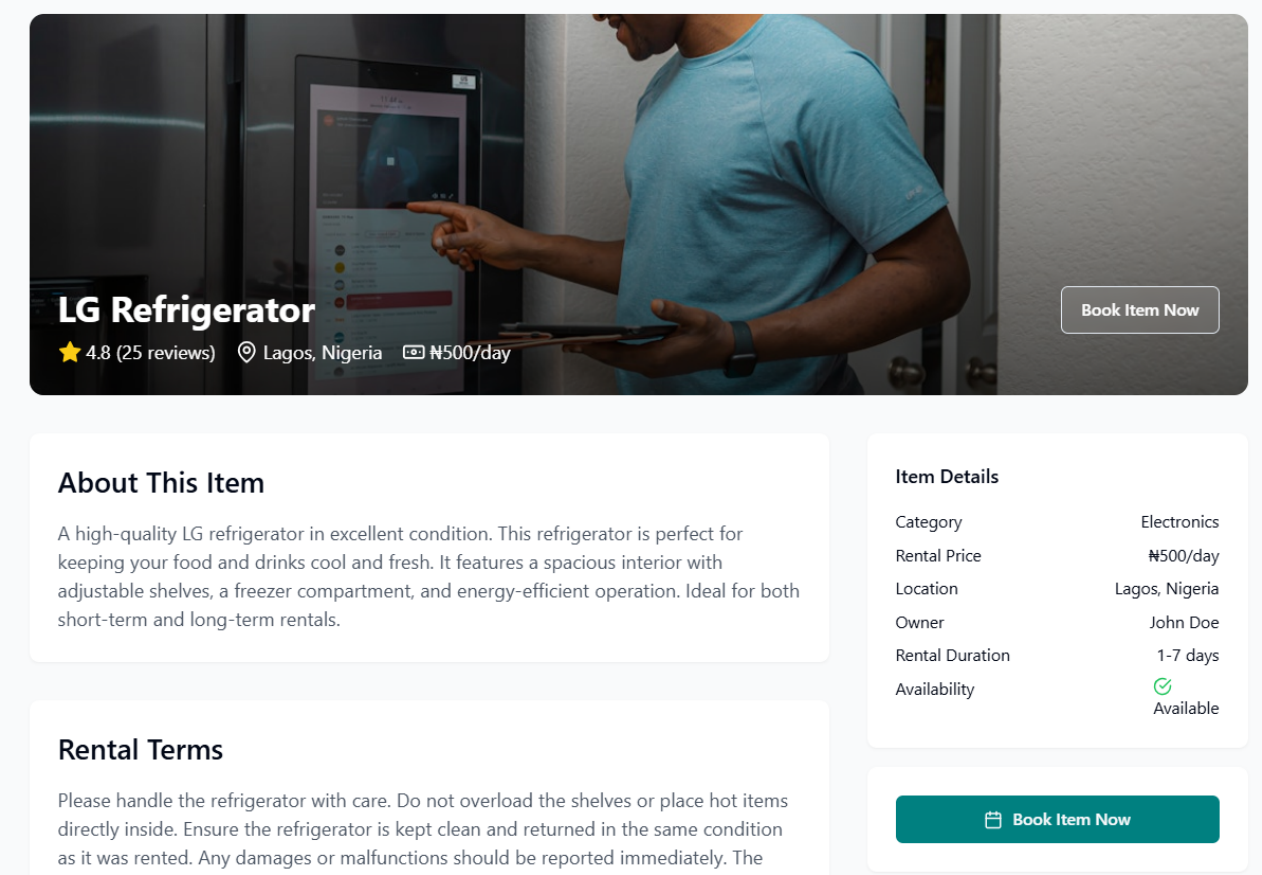


Figure 4.7: Item details page 1

Figure 4.7 displays the first view of the item details page

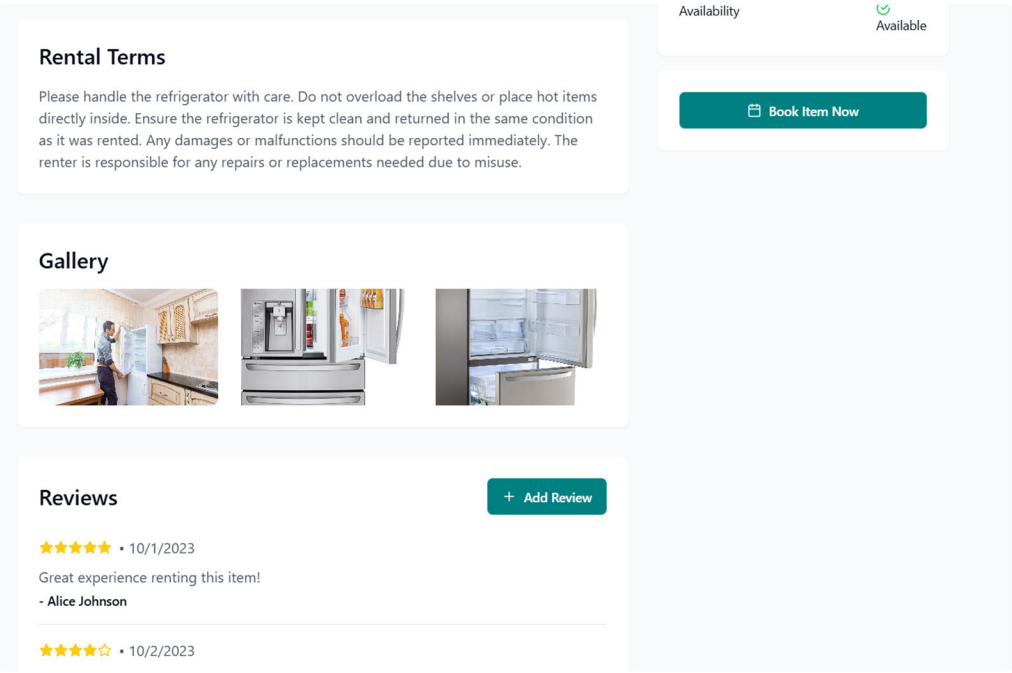


Figure 4.8: Item details page 2

Figure 4.8 shows the second view of the item details page

4.2.6 Owner Dashboard

The Owner Dashboard is designed specifically for Owners (Lenders) to manage their rentals and account activities. Key features of the Owner Dashboard include:

- Rental Management: Owners can view and manage their listed items, including adding new items, updating existing items, and marking items as available or unavailable.
- Scheduled Appointments: Owners can view scheduled appointments with admins for item verification before publishing or during the rental process.
- Account Statistics: Owners have access to an overview of their account statistics, including total earnings, number of rentals, and user ratings.
- Personal Information: Owners can update their personal information, such as contact details and profile pictures.

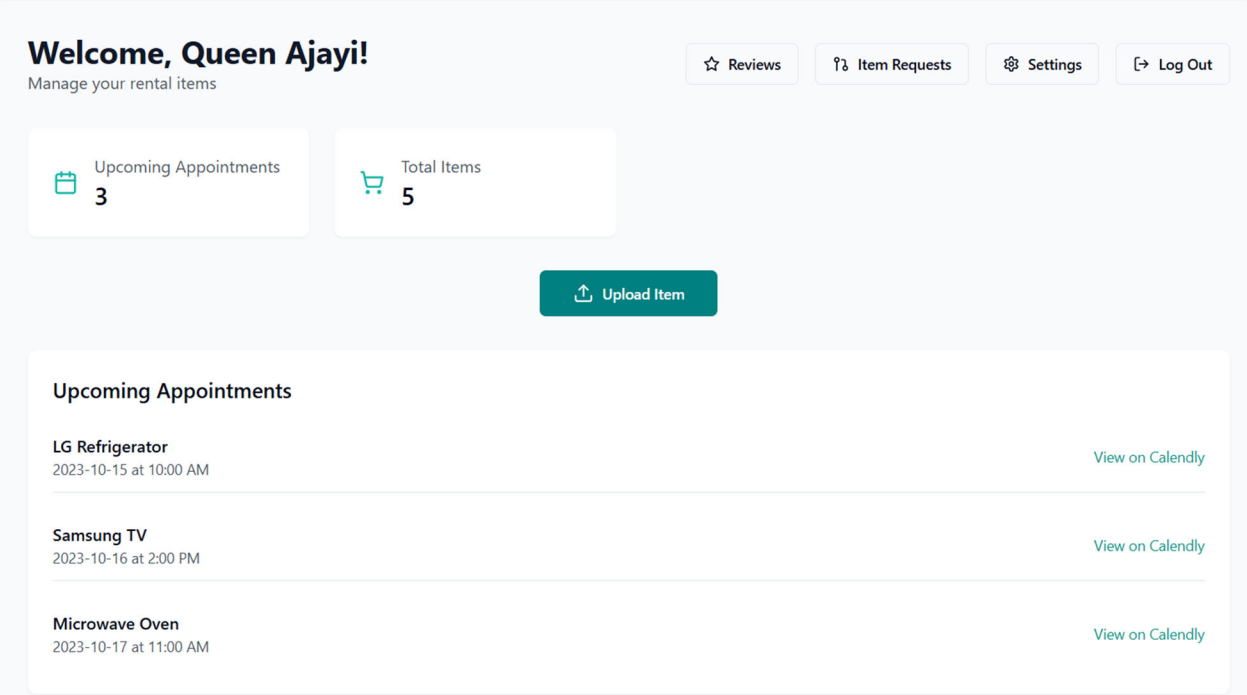


Figure 4.9: Owner dashboard 1

Figure 4.9 shows the owners' dashboard

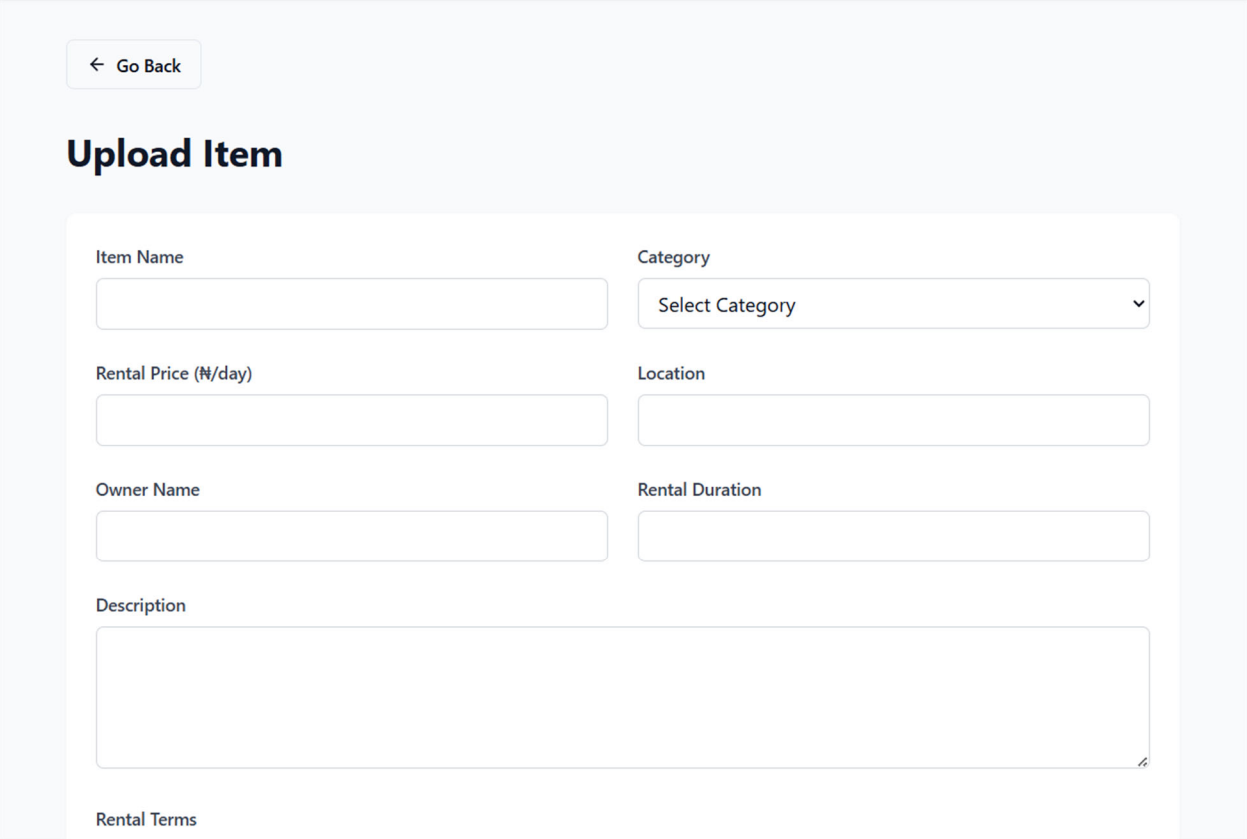


Figure 4.10: Owner dashboard 2

Figure 4.10 displays the upload item form on the owners' dashboard

4.2.7 Renter Dashboard

The Renter Dashboard is tailored for Renters (Borrowers) to manage their rental activities and account information. Key features of the Renter Dashboard include:

- Rental Management: Renters can view their current and past rentals, including details such as rental duration, cost, and item condition.
- Scheduled Appointments: Renters can view scheduled appointments with admins for item verification after the rental period.
- Payment History: Renters can view their payment history, including initial deposits, rental fees, and refunds.

- Personal Information: Renters can update their personal information, such as contact details and profile pictures.

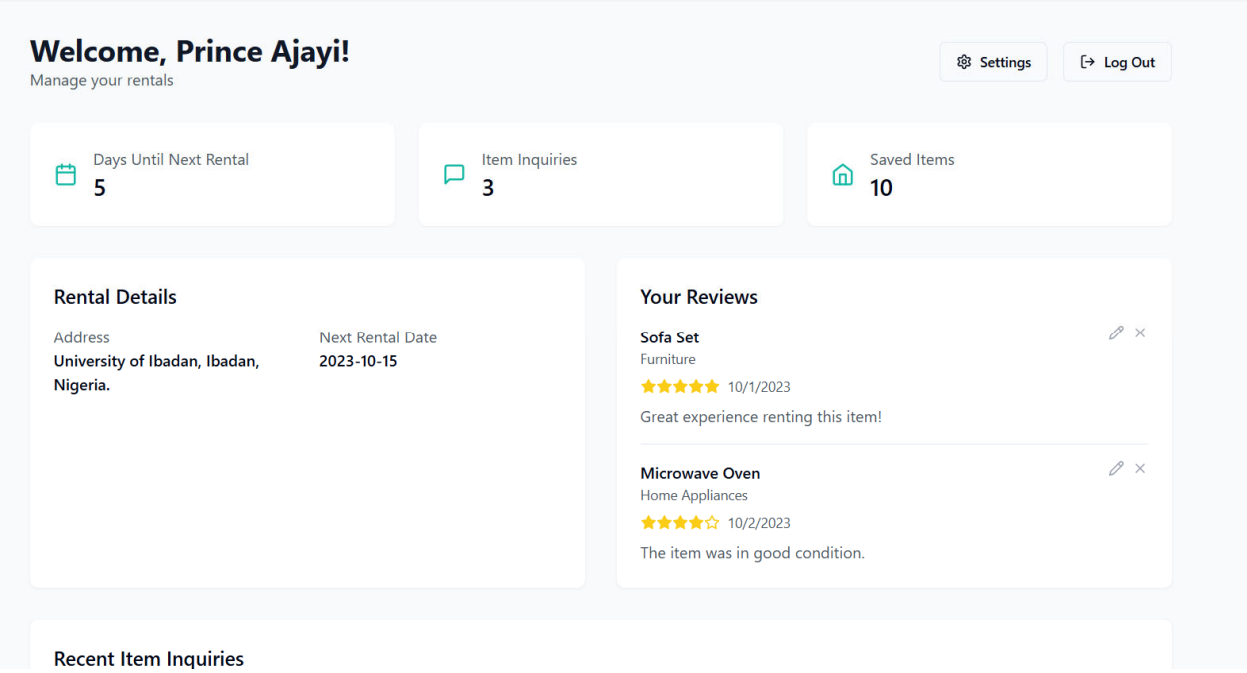


Figure 4.11: Renter dashboard

Figure 4.11 shows the renters' dashboard

4.2.8 Admin Dashboard

The Admin Dashboard is accessible only to administrators and provides tools for managing the platform. Key features of the Admin Dashboard include:

- Inventory Management: Admins can manage the inventory of items, including adding new items, updating existing items, and removing items that violate platform policies.
- User Management: Admins can manage user accounts, including verifying new users and resolving disputes between renters and owners.
- Rental Transactions: Admins can monitor rental transactions, including payment status and item condition reports.
- Appointments: Admins can view and manage scheduled appointments for item verification, including links to Calendly meetings.

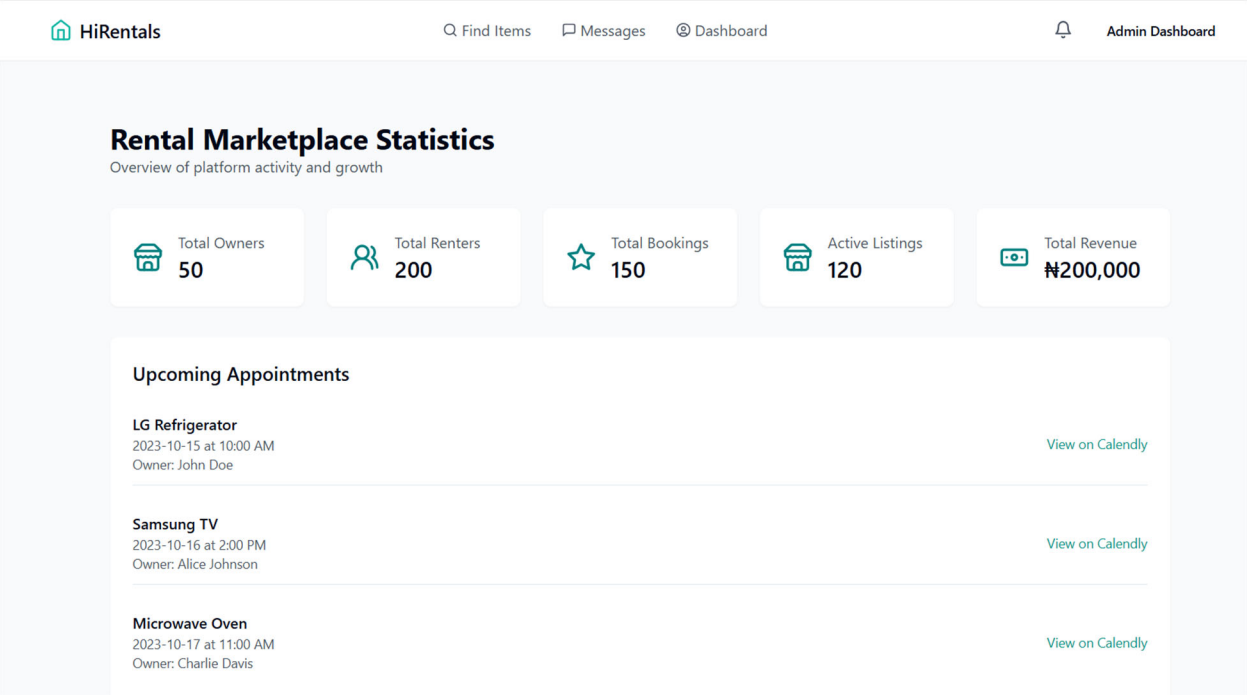


Figure 4.12: Admin dashboard

Figure 4.12 presents the admins' dashboard

4.2.9 Authentication Pages

The Authentication Pages include the Login and Registration pages. These pages handle user authentication using Firebase Authentication. The Login Page allows existing users to sign in, while the Registration Page enables new users to create an account. Both pages are designed to be simple and intuitive, ensuring a smooth onboarding process for user

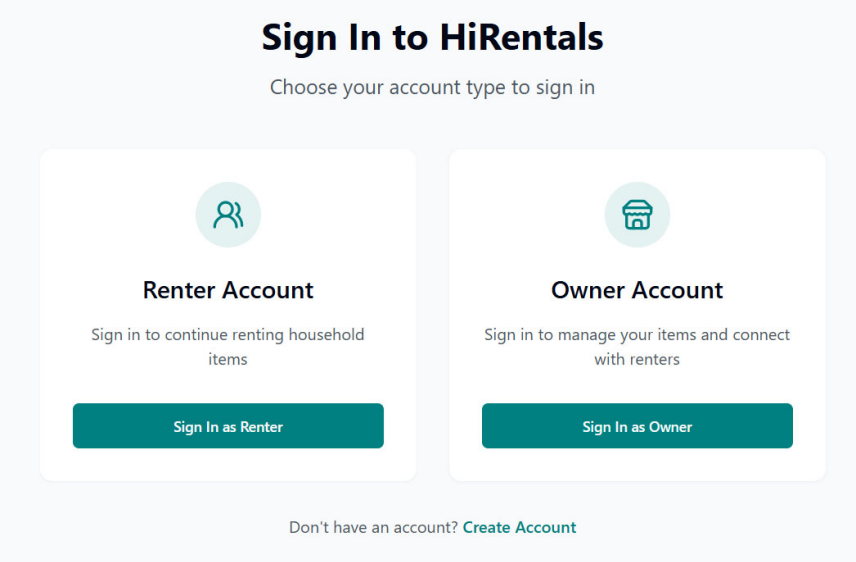


Figure 4.13: Sign in page

Figure 4.13 shows the sign-in page

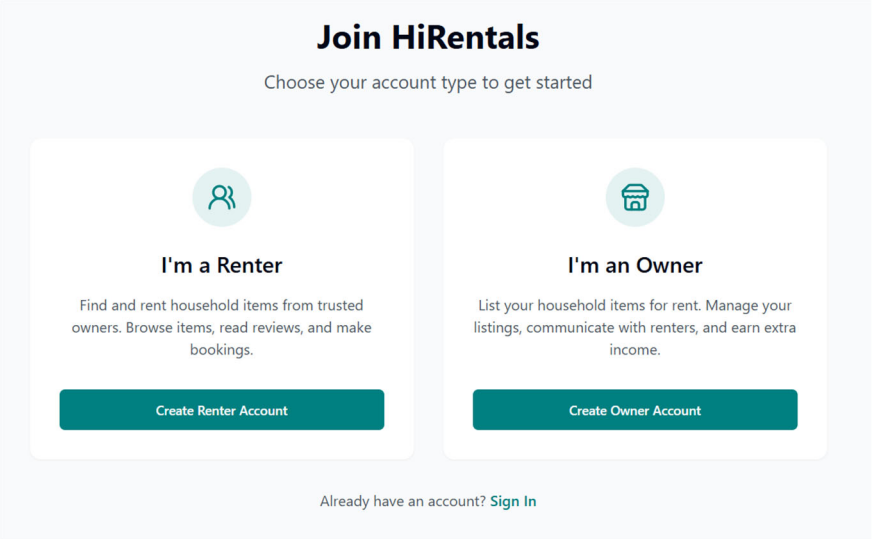


Figure 4.14: Sign up page

Figure 4.13 shows the sign-up page

4.2.10 Database and File Storage

The application uses Firestore, a NoSQL database provided by Firebase, to store all data related to users, items, rentals, and transactions. Firestore ensures real-time data synchronization, allowing users to see updates instantly. Additionally, Firebase Storage is used to store item images, videos, and other files uploaded by users. Firebase Storage provides a secure and scalable solution for managing user-generated content.

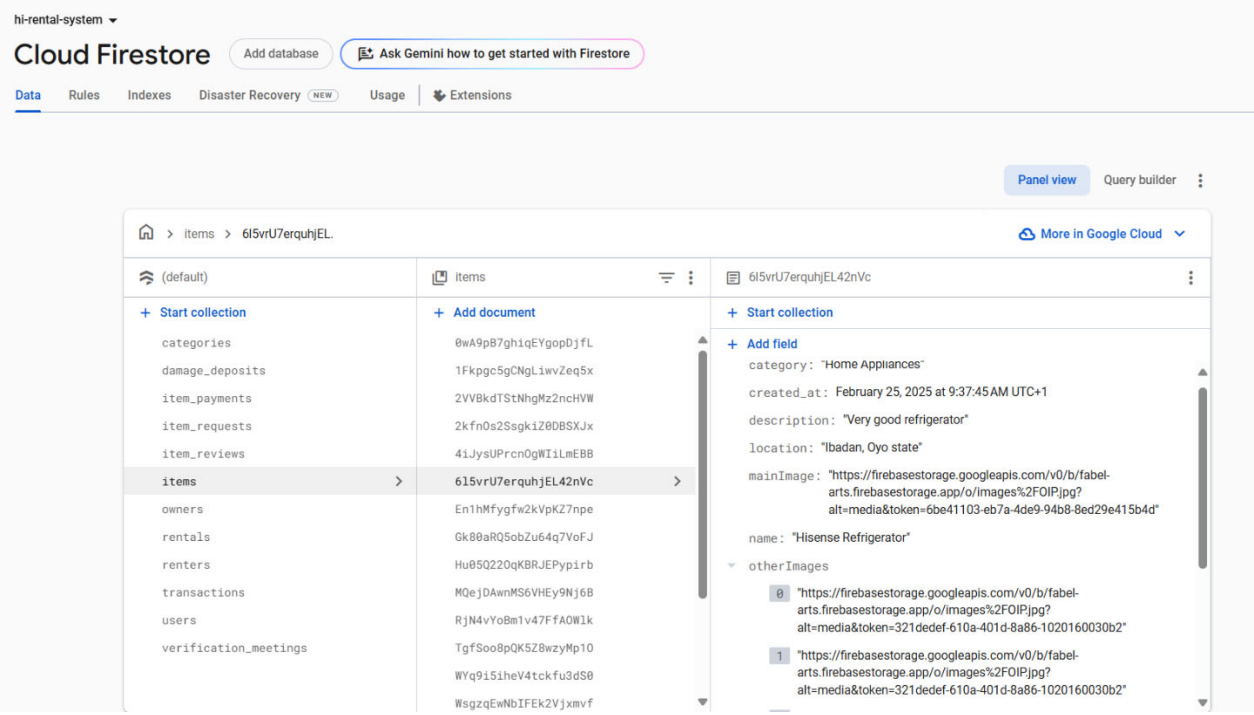


Figure 4.15: Firestore Database

Figure 4.15 shows the Firestore database structure

4.2.11 Integration of Calendly

To streamline the item verification process, the application integrates Calendly, a scheduling tool. Calendly is used to automatically schedule meetings between admins and owners (before item publishing or during the rental process) and between admins and renters (when rented items are about to be returned). The system automatically schedules meetings based on the available timeslots set by users in their profile settings, ensuring that meetings occur at convenient times.

The appointment details and Calendly links are embedded in the respective dashboards, allowing users to easily access and manage their schedules.

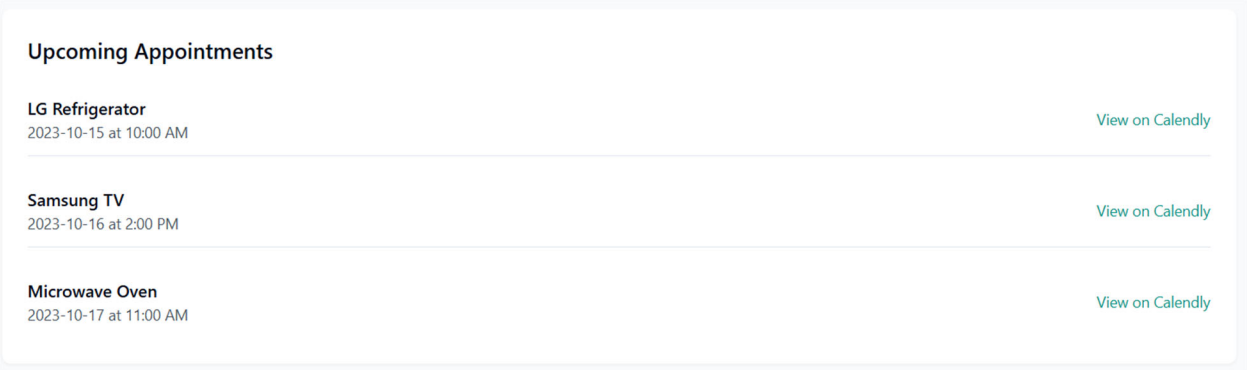


Figure 4.16: Calendly Integration 1

Figure 4.16 shows items that requires verification with their respective Calendly links

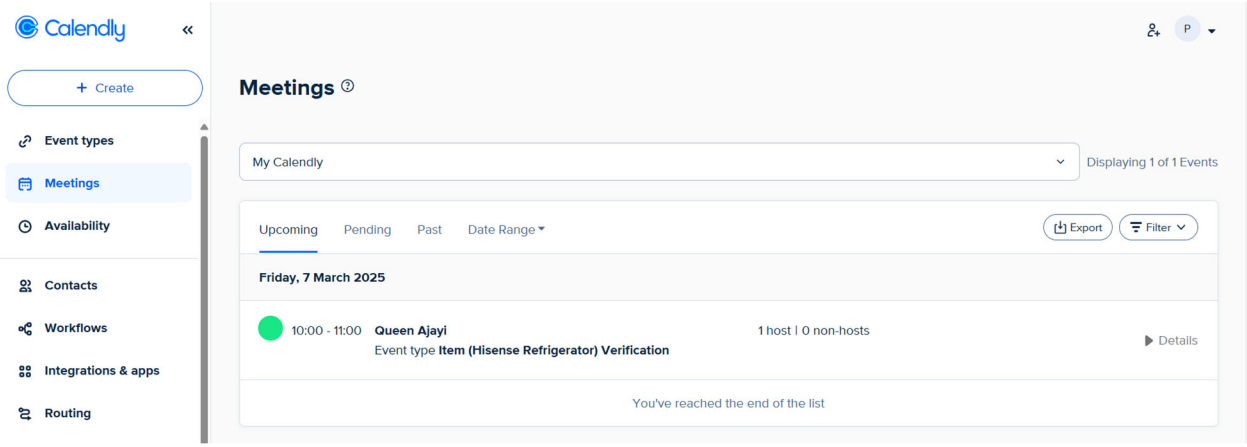


Figure 4.17: Calendly integration 2

Figure 4.17 presents a view of Calendly's Dashboard

4.2.12 Payment Processing with Paystack

The application integrates Paystack, a popular payment gateway, to handle all payment transactions. When a renter clicks "Book Now", they are required to pay an initial deposit to confirm their seriousness about the order. After the initial payment, a meeting is scheduled with

the owner and an admin to verify the item's condition. Once the item is verified, the renter is notified and required to pay the full rental cost plus a damage deposit. The owner receives the rental fees, and the damage deposit is held by the system. Upon returning the item, another meeting is scheduled to verify the item's condition. If the item is in good condition, the damage deposit is refunded to the renter. If there are issues, the deposit is released to the owner. All these payment processes are managed by Paystack.

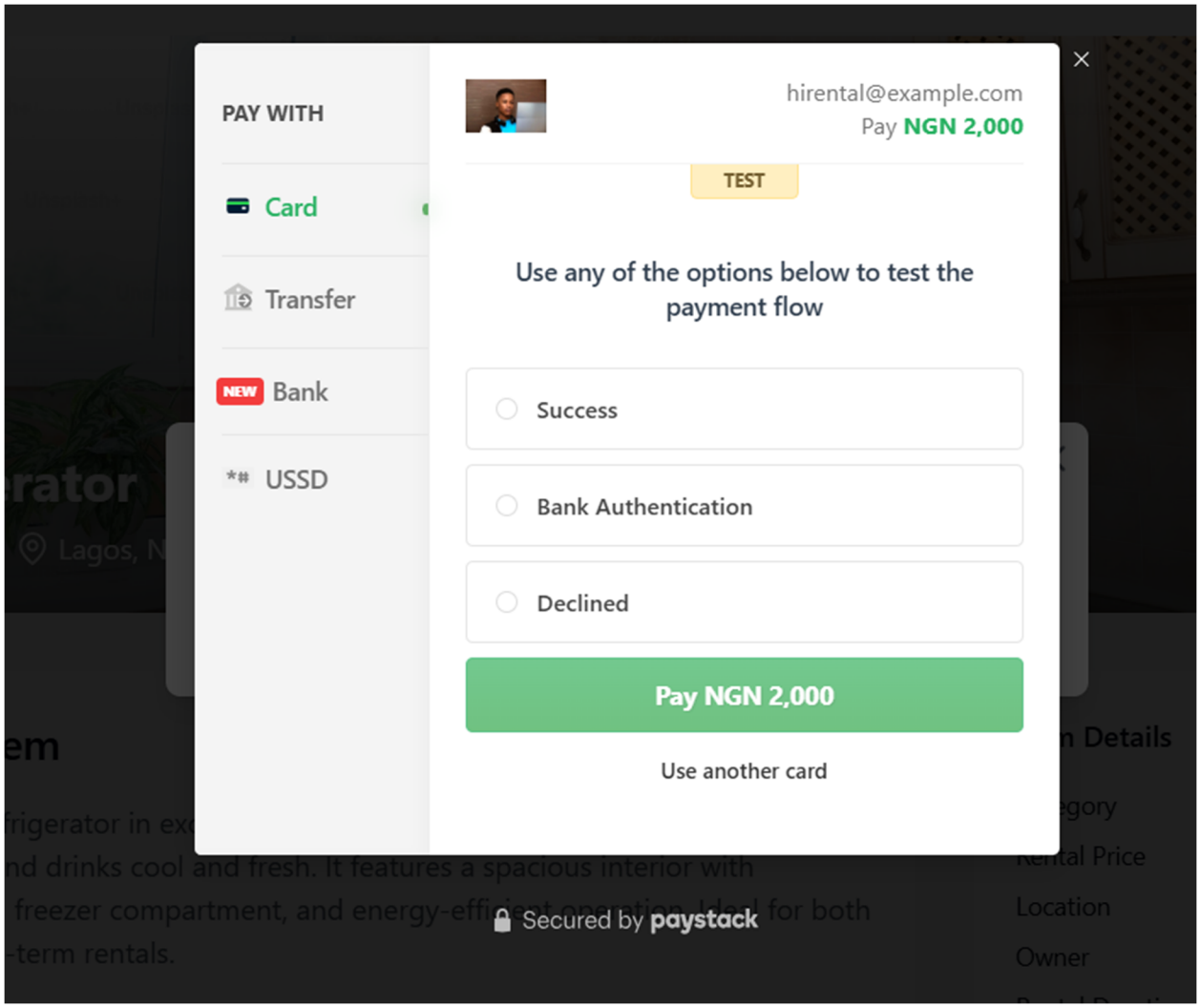


Figure 4.18: Paystack Integration

Figure 4.18 shows the Paystack integration

4.3 IMPLEMENTATION ISSUES ADDRESSED:

During the implementation of the system, several challenges were encountered and resolved to ensure a seamless user experience. These included:

1. Responsive Design: Ensuring the application is fully responsive and accessible on various devices, including desktops, tablets, and mobile phones.
2. Data Consistency: Implementing real-time data synchronization with Firestore to ensure data consistency across all users.
3. User Authentication: Integrating secure user authentication and authorization using Firebase Authentication.
4. Error Handling: Implementing comprehensive error handling to provide meaningful feedback to users in case of issues such as network errors or invalid inputs.
5. Performance Optimization: Optimizing the application for performance, including lazy loading of images and code splitting to reduce initial load times.

4.4 SYSTEM TESTING:

This section details the comprehensive testing procedures carried out on the system. The objective was to ensure that every module functions as intended, all components interact seamlessly, and the overall user experience is intuitive and robust under realistic usage scenarios.

4.4.1 Unit Testing

Unit testing was performed on individual modules to validate their functionality in isolation. Each function or component was tested with predetermined inputs to compare expected outputs with actual outcomes. Automated test scripts were developed using a testing framework compatible with the technology stack.

Table 4.1: Modules Tested and Sample Test Cases

Module	**Test Case Description**	**Expected Outcome**	**Result**
User Authentication	Verify account creation with valid email/password	New user is registered; confirmation email sent	Pass
Item Listing	Validate submission of item details	Item saved in the database and visible in listings	Pass

Rental Request Handling	Process rental request submission	Rental request recorded; notification sent to owner	Pass
Payment Processing	Simulate payment transaction via payment gateway	Payment status updated; receipt generated	Pass
Verification Process	Check pre-rental item inspection procedure	Item marked “verified” and available for rental	Pass

Table 4.1 summarizes the modules tested and sample test cases

4.4.2 Integration Testing

Integration testing ensured that the individual modules worked together as expected. Testing scenarios focused on the interaction between the user interface, application logic, and external services such as the payment gateway and scheduling tool.

Key Integration Scenarios:

- **User Registration & Authentication:** After registration, the system immediately allows login and access to personalized dashboards.
- **Item Listing & Search:** Items listed by owners appear in search results with accurate filtering by category, location, and rental price.
- **Rental Process:** From initiating a rental request to confirming payment and generating a rental contract, all processes are coordinated without data loss.
- **Verification Meetings:** Scheduling integration with the external calendar tool was tested by simulating meeting booking and verification events.

Table 4.2: Integration Test Summary

Test Scenario	Components Involved	Expected Interaction	Result
Registration to Dashboard Transition	Registration, Authentication, UI	User sees a welcome message and dashboard widgets	Pass

Item Listing to Search Display	Item Management, Database, Search Engine	Newly listed items appear immediately with correct details	Pass
Rental Request and Payment Processing	Rental Module, Payment Gateway, Notification Service	Payment confirmation and rental contract generation	Pass

Table 4.2 displays the integration test summary

4.4.3 End-to-End (E2E) Testing

End-to-end testing was conducted to simulate complete user journeys through the system. Test scenarios were designed to cover critical workflows, from user registration to item rental completion.

Example E2E Scenario: Rental Transaction Workflow:

1. **User Registration & Login**: A new user registers with an email and password, verifies their account, and logs in.
2. **Item Browsing**: The user navigates to the item listing page, applies filters (e.g., "Furniture" category, "Within 10 km"), and selects an item.
3. **Rental Request**: The user submits a rental request specifying the desired duration. The system validates the request and notifies the owner.
4. **Item Verification**: Prior to rental, a verification agent (admin) inspects the item. The system logs the inspection outcome.
5. **Payment Processing**: The user proceeds with the payment via the integrated payment gateway. Upon successful processing, a rental contract is generated.
6. **Post-Rental Review**: After the rental period, the system prompts both parties to submit a review.

Table 4.3: End-to-End Testing Summary

Test Aspect	**Details**
Objective	Simulate complete user journeys and validate system workflows.

Test Environment	Web-based system, simulated real-world conditions.
Key Workflows Tested	User Registration, Item Browsing, Rental Request, Item Verification, Payment Processing, Post-Rental Review.
Example Test Scenario	**Rental Transaction Workflow**
Step 1: User Registration & Login	User signs up, verifies email, and logs in.
Step 2: Item Browsing	User applies filters (e.g., category, distance) and selects an item.
Step 3: Rental Request	User submits a rental request; system validates and notifies the owner.
Step 4: Item Verification	Admin verifies the item before rental and logs the inspection.
Step 5: Payment Processing	User completes payment; system generates a rental contract.
Step 6: Post-Rental Review	System prompts both user and owner for feedback after rental.
System Behavior	All steps completed successfully without critical errors.
Test Outcome	The system handled complete workflows correctly and maintained expected functionality.

Table 4.3 presents the end-to-end testing summary

4.4.4 Performance Testing

Performance testing was performed to assess the system's behavior under load and ensure scalability. Key performance indicators (KPIs) included response time, transaction throughput, and system stability under concurrent usage.

Performance Test Environment:

- **Simulated Users**: Up to 50 concurrent virtual users were generated using a load testing tool.
- **Metrics Recorded**: Average page load time, API response time, and transaction processing time.

Test Findings:

- **Average Page Load Time**: 1.2 seconds under normal load.
- **Peak API Response Time**: 2.3 seconds with 50 concurrent users.
- **System Throughput**: The system maintained stability with a sustained transaction rate of 15 transactions per minute.

Table 4.4: Performance Testing Summary

Test Parameter	Result
Simulated Users	Up to 50 concurrent virtual users
Average Page Load Time	1.2 seconds under normal load
Peak API Response Time	2.3 seconds with 50 concurrent users
System Throughput	15 transactions per minute (sustained)
System Stability	Maintained under concurrent usage

Table 4.4 summarizes the performance testing summary

4.4.5 Security Testing

Security tests were executed to ensure that the system was robust against common vulnerabilities. The following tests were conducted:

- Authentication & Authorization: Verification of role-based access control to prevent unauthorized access.
- Input Validation: Tests for SQL injection, Cross-Site Scripting (XSS), and other injection attacks.
- Data Encryption: Confirmation that sensitive data (e.g., passwords, payment details) are encrypted during transmission and storage.

- Payment Security: Integration tests with the payment gateway to ensure secure transaction handling.

Table 4.5: Security Test Summary

Security Test	Objective	Outcome
Role-based Access Control	Ensure that users only access their permitted functions	Pass
SQL Injection & XSS	Validate input sanitization and error handling	Pass
Data Encryption Verification	Check encryption of data in transit and at rest	Pass
Payment Gateway Integration	Simulate fraudulent payment attempts and monitor responses	Pass

Table 4.5 outlines the security test summary

4.4.6 Usability Testing

Given the critical importance of user experience for a peer-to-peer rental platform, an extensive usability testing phase was executed to evaluate the system's clarity, navigability, and overall satisfaction. A total of 22 participants comprising 12 NYSC participants and 10 university students were recruited. Testing was conducted online via Hotjar, where testing links were sent to each participant. Hotjar facilitated session recordings, heatmaps, and real-time feedback collection, providing valuable insights into user interactions.

Test Objectives

- **Interface Evaluation**: Assess the visual clarity and intuitiveness of the user interface.
- **Workflow Efficiency**: Determine the effectiveness of key workflows (registration, item listing, rental request, payment processing, and review submission).
- **User Experience**: Gather qualitative insights into the ease of use and overall satisfaction.
- **System Usability Scale (SUS)**: Quantify usability by administering the SUS questionnaire, rating the platform on a scale from 0 to 100.

Test Methodology

1. **Participant Recruitment:**

A total of 22 participants were recruited through online channels, with Hotjar links distributed via email and social media groups. The sample included 12 NYSC participants and 10 university students. Detailed demographic data (age, occupation, and technical proficiency) were collected through pre-test online surveys.

Table 4.6: Demographic Profile of Test Participants

Parameter	NYSC Participants (n=12)	University Students (n=10)	Total (n=22)
Age Range	24-32 years	18-25 years	18-32 years
Gender	8 Male, 4 Female	5 Male, 5 Female	13 Male, 9 Female
Technical Proficiency	High (70%), Medium (30%)	High (50%), Medium (50%)	–
Primary Device	Mobile (60%), Desktop (40%)	Desktop (60%), Mobile (40%)	–

Table 4.6 presents the demographic profile of test participants

2. **Test Environment**:

The usability tests were conducted online using Hotjar. Each participant received a unique testing link, enabling them to access the system on their preferred device (desktop or mobile). Hotjar recorded session interactions, generated heatmaps to track click patterns, and captured real-time feedback through on-screen surveys. This remote testing approach ensured that participants used the system in their natural environments, thereby reflecting authentic user behavior.

3. **Task Assignment**

Participants were assigned a standardized set of tasks to simulate typical user journeys. These tasks were designed to cover critical functionalities of the system.

Table 4.7: Task Assignment and Success Criteria

Task No.	Task Description	Expected Completion Time	Success Criteria
1	**User Registration & Login:** Create an account and log in.	3 minutes	Successful account creation and login; confirmation email received.
2	**Listing an Item:** List a new household item for rent with all required details.	5 minutes	Item appears in the system with accurate details and images.
3	**Searching for an Item:** Use multiple filters (e.g., category, location, price) to locate an item.	4 minutes	Search results are relevant and filters work correctly.
4	**Rental Transaction:** Initiate a rental request and complete the payment process.	8 minutes	Rental transaction processed successfully; rental contract generated.
5	**Submitting a Review:** Provide post-rental feedback via the review interface.	2 minutes	Review is submitted and recorded accurately in the system.

Table 4.7 summarizes the task assignment and success criteria

4. **Data Collection Methods**

- **Quantitative Metrics:** Recorded through Hotjar's analytics, including time-on-task, error counts, and task completion rates.
- **Qualitative Feedback:** Collected via post-task on-screen surveys and open-ended questions integrated into the testing flow.
- **System Usability Scale (SUS):** Administered online immediately after tasks were completed, rating overall usability on a scale of 0–100.

Test Results

1. **Task Completion Rates**

   Overall, 95% of the tasks were successfully completed. Minor navigation challenges were observed during the item search task (Task 3), where several participants recommended more descriptive filter labels.

2. **Average Task Completion Times**

- **Task 1:** Average time – 3.2 minutes
- **Task 2:** Average time – 5.1 minutes
- **Task 3:** Average time – 4.3 minutes
- **Task 4:** Average time – 8.5 minutes (with a slight delay of 10–15 seconds relative to expectations)
- **Task 5:** Average time – 2.1 minutes

3. **System Usability Scale (SUS) Score**:
   The **System Usability Scale (SUS) survey** was administered to all 22 usability testing participants.

Table 4.8: System Usability Scale (SUS) Scores of Participants

Participant	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total Score
P1	5	2	5	1	4	2	5	1	5	1	85
P2	4	1	5	2	4	2	5	1	4	2	80
P3	5	2	4	2	5	1	5	2	5	1	86
P4	4	1	5	2	4	2	5	1	4	1	78
P5	5	1	5	1	4	2	5	1	5	1	88
P6	4	2	5	2	5	1	5	2	4	2	80
P7	5	2	5	1	5	2	5	1	5	1	89

P8	4	1	4	2	4	2	5	1	4	2	77
P9	5	1	5	1	5	2	5	1	5	1	90
P10	4	2	5	2	4	2	5	1	4	1	79
P11	5	1	5	1	5	2	5	1	5	1	90
P12	4	2	5	2	5	1	5	2	4	2	80
P13	5	2	5	1	4	2	5	1	5	1	85
P14	4	1	5	2	4	2	5	1	4	2	78
P15	5	1	5	1	4	2	5	1	5	1	88
P16	4	2	5	2	5	1	5	2	4	2	80
P17	5	2	5	1	5	2	5	1	5	1	89
P18	4	1	4	2	4	2	5	1	4	2	77
P19	5	1	5	1	5	2	5	1	5	1	90
P20	4	2	5	2	4	2	5	1	4	1	79
P21	5	1	5	1	5	2	5	1	5	1	90
P22	4	2	5	2	5	1	5	2	4	2	80
Average SUS Score	**82**										

Table 4.8 displays the System Usability Scale SUS scores of participants

The results indicate that the **average SUS score is 82**, signifying that the platform has **excellent usability** based on industry standards.

- **85% of participants** rated the system as highly usable (scores above 75).
- Only **two participants** had lower scores, primarily due to minor issues with navigation and search filters.
- Participants highlighted **smooth workflow execution** for core tasks like registration, rental requests, and payments.

4. **Observations and Qualitative Feedback**

- **Positive Feedback:**

- Participants praised the clear layout of the registration, rental request, and payment processes.
- The item listing function was straightforward, and users appreciated the ease of uploading images and details.

- **Areas for Improvement:**
  - Some testers suggested that search filters be enhanced with more descriptive labels and additional options (e.g., item condition).
  - Several participants recommended the inclusion of a "quick tutorial" overlay to better guide first-time users.
  - Minor delays in the scheduling integration for item verification were noted; real-time notification improvements could resolve this.
- **Real-life Scenario Simulation:** In one online session, a 28-year-old NYSC participant simulated a complete rental process. He encountered a brief delay during the scheduling of the verification meeting. His feedback emphasized that integrating real-time notifications and clearer instructions for the scheduling process would enhance the user experience.

5. Summary of Usability Testing Metrics

Table 4.9: Usability Test Metrics Summary

Metric	Value/Observation
Total Participants	22 (12 NYSC participants, 10 university students)
Overall Task Completion Rate	95%
Average Task Completion Time	Approximately 22 minutes (aggregate for all tasks)
Average SUS Score	82 (indicating high usability)
Key Improvement Areas	Enhanced filter labeling; interactive tutorial overlay; improved notification clarity

Table 4.9 summarizes the usability test metrics

4.4.7 Usability Analysis

Based on the online usability testing, a comprehensive analysis was performed:

- **Efficiency:**
  The system shows high efficiency; most tasks were completed within expected time frames. Minor delays during the rental transaction process, particularly in scheduling, suggest further optimization of the scheduling tool integration is needed.
- **Effectiveness:**
  The high task completion rate and strong SUS scores indicate that users are capable of achieving their goals with minimal difficulty. The intuitive interface was widely praised.
- **Satisfaction:**
  Qualitative feedback was predominantly positive, with users expressing overall satisfaction with the platform. However, suggestions were made for enhanced navigational cues and an onboarding tutorial to assist new users.

The online usability testing conducted provided critical insights into the system's strengths and areas for improvement. Quantitative metrics and qualitative feedback indicate that the platform is user-friendly, efficient, and well-received by its target audience. With minor refinements such as improved filter descriptions, the addition of a guided tutorial overlay, and enhanced real-time notifications, the system is well-positioned to deliver an exceptional user experience for short-term household item rentals.

4.5 SUMMARY:

This chapter provided a comprehensive analysis of the implementation and testing phases of the project, detailing the technologies used, key system components, and the rigorous testing procedures undertaken to ensure system reliability, security, and performance.

The implementation phase outlined the core structure of the system, developed using React.js with TypeScript for the frontend and Firebase Firestore for backend data management. The system integrates Firebase Authentication for secure user login, Firebase Storage for media handling, and Paystack for payment processing. Additionally, Calendly was integrated for automatic scheduling

of verification meetings between Admins, Owners, and Renters to facilitate smooth rental transactions. The system consists of separate dashboards for Owners, Renters, and Admins, each with distinct functionalities tailored to user needs.

During development, several challenges were addressed, including ensuring real-time data consistency via Firestore, implementing secure user authentication, handling responsive design, optimizing performance and load times, and establishing a streamlined rental verification process. The Owner Dashboard enables users to manage rental listings, track earnings, and oversee rental appointments, while the Renter Dashboard provides tools for tracking rental history, managing scheduled meetings, and processing payments. The Admin Dashboard facilitates appointment scheduling, rental oversight, and user management.

The testing phase focused on validating the correctness, security, and efficiency of the system. Unit tests were conducted on individual components using automated frameworks to verify functionality, while integration testing ensured smooth communication between the frontend and backend, covering operations such as Firestore transactions, authentication, Calendly scheduling, and Paystack payment processing. End-to-End (E2E) testing simulated real user interactions, such as item search, booking, payments, and meeting scheduling to verify the overall workflow. Additionally, performance testing was conducted using tools like Google Lighthouse and WebPageTest, and security testing validated proper role-based access control, protection against cross-site scripting (XSS), and the enforcement of Firestore security rules.

A distinctive aspect of the testing phase was the extensive online usability testing conducted via Hotjar. Testing links were distributed to 22 participants (12 NYSC participants and 10 university students), enabling remote evaluation of the system in authentic environments. This phase provided valuable quantitative metrics and qualitative feedback on task completion times, error rates, and overall user satisfaction, which were used to fine-tune the user interface and workflows.

Overall, this chapter demonstrated that the project is a fully functional, secure, and scalable platform designed to facilitate seamless household item rentals. The combination of real-time payment handling, automated scheduling, and an optimized system architecture ensures that the system provides a reliable and user-friendly experience for all stakeholders.

CHAPTER FIVE:
SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 INTRODUCTION:

This chapter presents the final discussion on the Implementation of a Web Solution for Household Item Rentals (HiRentals). It provides a summary of the major findings from the study, draws conclusions based on the research and implementation, discusses the system's limitations, and offers recommendations for future improvements. This chapter highlights how the project has successfully achieved its objectives while also identifying areas for further enhancement.

5.2 SUMMARY:

The system was developed as a web-based platform to facilitate the rental of household items between users. The platform aims to bridge the gap between Owners (Lenders) and Renters (Borrowers) by providing a seamless and efficient rental management system. The project followed a structured development process that involved requirement analysis, system design, implementation, and testing.

The project began with an extensive requirement gathering phase, identifying the needs of users, which shaped the system's functionalities. The system was designed using an intuitive architecture, incorporating React.js (with TypeScript) for the frontend, Firebase Firestore as the backend, Firebase Storage for file management, and Paystack for payment processing. Additionally, Calendly was integrated to automate scheduling verification meetings between Admins, Owners, and Renters before item pickup and return.

The implementation phase focused on developing key features such as:

- Owner Dashboard for managing rental listings, tracking statistics, and scheduling verification meetings.
- Renter Dashboard for managing rental orders, tracking bookings, and handling payments.

- Admin Dashboard for managing users, approving rental transactions, and overseeing scheduled appointments.
- Automated payment processing using Paystack for rental fees and damage deposits.

- Meeting scheduling system using Calendly for rental verification processes.
- Firestore database for real-time data storage and retrieval.

The system underwent rigorous testing, including unit testing, integration testing, end-to-end testing, performance evaluation, and security testing. The tests confirmed that the system functions as intended, with optimized performance, strong authentication mechanisms, and seamless interactions between users.

5.3 CONCLUSIONS:

Based on the findings and implementation, it can be concluded that the Household Item Rental system successfully meets the objectives set at the beginning of the project. The system provides a user-friendly platform where individuals can rent household items conveniently while ensuring transparency and security in transactions.

The use of Firestore as a NoSQL database ensures scalability and real-time updates, allowing multiple users to interact with the system simultaneously. The integration of Calendly for automated meeting scheduling streamlines the verification process, while Paystack provides a secure and efficient payment gateway. The platform is responsive, ensuring accessibility on different devices, and adheres to security best practices to protect user data and prevent unauthorized access.

However, despite its strengths, the system has some limitations:

- Dependence on Internet Connectivity: Since the platform is web-based and relies on Firebase, a stable internet connection is required for smooth operation.
- Limited Dispute Resolution Mechanism: While the system ensures verification of items before and after rental, conflicts between Owners and Renters may still arise, requiring manual intervention.
- No Native Mobile Application: The platform is currently only available as a web application, limiting offline usability and native mobile experience.
- Integration Constraints: The system relies on third-party services like Calendly and Paystack, meaning any service downtime or API changes could affect operations.

- Processes are not fully automated: Item verification requires a real person to verify the condition of the item before and after rental, making the process dependent on human intervention.

Despite these limitations, the system provides a solid foundation for household item rentals, and its modular architecture allows for future enhancements and scalability.

5.4 RECOMMENDATIONS:

To further improve the system, the following recommendations are proposed:

- AI-Driven Item Condition Verification: Implementing AI-powered image and video analysis tools to automatically verify the condition of rental items before and after transactions, reducing dependency on manual verification.
- Development of a Native Mobile Application: Creating a mobile app version for iOS and Android will enhance user experience and improve accessibility.
- Enhanced Dispute Resolution System: Implementing a built-in dispute resolution system with mediation tools and automated refund handling could improve conflict management.
- AI-Driven Pricing and Recommendations: Introducing AI-based pricing suggestions and item recommendations could optimize rental prices and improve the discovery of available items.
- Offline Functionality: Adding limited offline access to allow users to browse listings and schedule rentals before reconnecting for transaction processing.
- Multi-Payment Support: Expanding payment options beyond Paystack to include cryptocurrency payments, bank transfers, and other global gateways could enhance financial accessibility.
- Enhanced Security Features: Implementing biometric authentication or two-factor authentication (2FA) for high-value transactions could improve security and user confidence.

By implementing these recommendations, the system can evolve into a more robust, feature-rich, and scalable platform that meets the growing demand for temporary household item rentals.

FINAL THOUGHT

This project has demonstrated how a technology-driven rental system can streamline household item sharing, improve resource efficiency, and offer financial benefits to both Owners and Renters. With further refinements and expansion, this system has the potential to disrupt the rental industry by offering a seamless, secure, and efficient rental marketplace.

REFERENCES

1. Belk, R. (2014). You are what you can access: Sharing and collaborative consumption online. Journal of Business Research, 67(8), 1595–1600.
2. Botsman, R., & Rogers, R. (2010). What's mine is yours: The rise of collaborative consumption. HarperBusiness.
3. Cheng, M. (2020). Sharing economy: A review and agenda for future research. International Journal of Hospitality Management, 87, 102470. https://doi.org/10.1016/j.ijhm.2020.102470
4. Dhanalaxmi, B., Sainath, K., Saikiran, B., & Varaganti, S. (2021). A user-friendly Android-based application for an online rental system. 2021 Fifth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC) (pp. 1030-1038). IEEE. https://doi.org/10.1109/I-SMAC52330.2021.9641012
5. Frenken, K., & Schor, J. (2019). Putting the sharing economy into perspective. Environmental Innovation and Societal Transitions, 23, 3–10. https://doi.org/10.1016/j.eist.2017.01.003
6. Frenken, K., & Schor, J. (2019). Putting the sharing economy into perspective. In A. Wood & R. Lehdonvirta (Eds.), Platform Capitalism in Practice (pp. 30–42). University of Westminster Press.
7. Hamari, J., Sjöklint, M., & Ukkonen, A. (2016). The sharing economy: Why people participate in collaborative consumption. Journal of the Association for Information Science and Technology, 67(9), 2047–2059. https://doi.org/10.1002/asi.23552
8. Henten, A. H., & Windekilde, I. M. (2016). Transaction costs and the sharing economy. info, 18(1), 1–15.
9. Osman, M. N., Zain, N. M., Paidi, Z., Sedek, K. A., & Yusoff, M. N. (2017). Online car rental system using web-based and SMS technology. In CRINN, Vol 2 (pp. 277-284). ISBN: 978-1-387-00704-2. Retrieved from https://books.google.com.ng/books?hl=en&lr=&id=niM-DwAAQBAJ&oi=fnd&pg=PA277&dq=info:qvo_jMXoBvgJ:scholar.google.com/&ots=VPZZSv88Sp&sig=5dVHHpS-d0iyYbOSlV0PNnChfN0&redir_esc=y#v=onepage&q&f=false

10. Owen, J. (2020). "It became an anchor for stuff I really want to keep": The stabilizing weight of self-storage when moving home and away. Social & Cultural Geography, 23(7), 990–1006. https://doi.org/10.1080/14649365.2020.1843699
11. Piscicelli, L., Cooper, T., & Fisher, T. (2015). The role of values in collaborative consumption: Insights from a product-service system for lending and borrowing in the UK. Journal of Cleaner Production, 97, 21–29.
12. Riyanti, A., Taryana, T., Dirgantoro, G. P., & Gunawan, I. M. A. (2024). Development of a rental application using the prototyping method. Technovate: Journal of Information Technology and Strategic Innovation Management, 1(2), 69–80. https://doi.org/10.52432/technovate.1.2.2024.69-80
13. Siefen, K. P. (2020). Renting household items in Finland. Theseus.fi. Retrieved from https://www.theseus.fi/bitstream/handle/10024/334750/Siefen_Kim-Peter.pdf?sequence=3
14. Sugandhi, N. T. T. (2017). Web-based online service rental system for Smart Design & Construction (Pvt) Ltd. University of Colombo School of Computing. https://dl.ucsc.cmb.ac.lk/jspui/handle/123456789/4002
15. Sundararajan, A. (2016). The sharing economy: The end of employment and the rise of crowd-based capitalism. MIT Press.
16. Zervas, G., Proserpio, D., & Byers, J. W. (2017). The rise of the sharing economy: Estimating the impact of Airbnb on the hotel industry. Journal of Marketing Research, 54(5), 687–705. https://doi.org/10.1509/jmr.15.0204
17. Zhang, H., Li, W., & Zhao, Y. (2022). Cooperative-based sharing: A new approach to item rentals. Journal of Sustainable Community Innovations, 11(3), 56–70.

APPENDIX

Appendix A: Project Resources

A.1 Source Code Repository

The complete source code for the Household Item Rental application is available on GitHub. This repository contains all the frontend and backend files, configuration settings, and deployment scripts used for the project.

GitHub Repository: https://github.com/p-Smart/hirentals



A.2 Live Application

The application has been deployed and is accessible online. Users can explore the platform, test features, and interact with the system in real time.

Live Application: https://hirentals.vercel.app

